

Introduction to Search and Propagation

Modern Constraint Solving
(ESSAI'26)

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In this mini-lecture...

Model-and-Solve → Constraint Programming

Search

Constraints and Propagators

Problem



Model

Problem

concrete

e.g., timetabling



Model

abstract

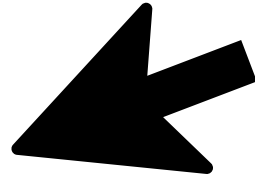
e.g., equations

unclear

Problem

concrete

e.g., timetabling



unambiguous

Model

abstract

e.g., equations

unclear

Problem

concrete

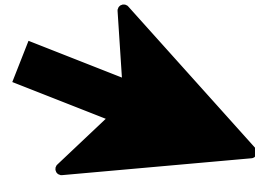
e.g., timetabling

unambiguous

Model

abstract

e.g., equations



Solve

algorithms

unclear

Problem

concrete

e.g., timetabling

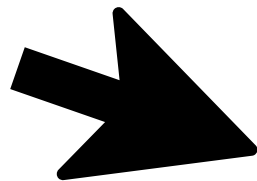
unambiguous

Model

abstract

e.g., equations

Our focus



Solve

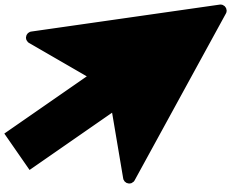
algorithms

Search

Branching + Inference/Propagation

Search

Branching + **Inference/Propagation**



Part 1: Propagation and Branching

Part 2: Conflict Analysis

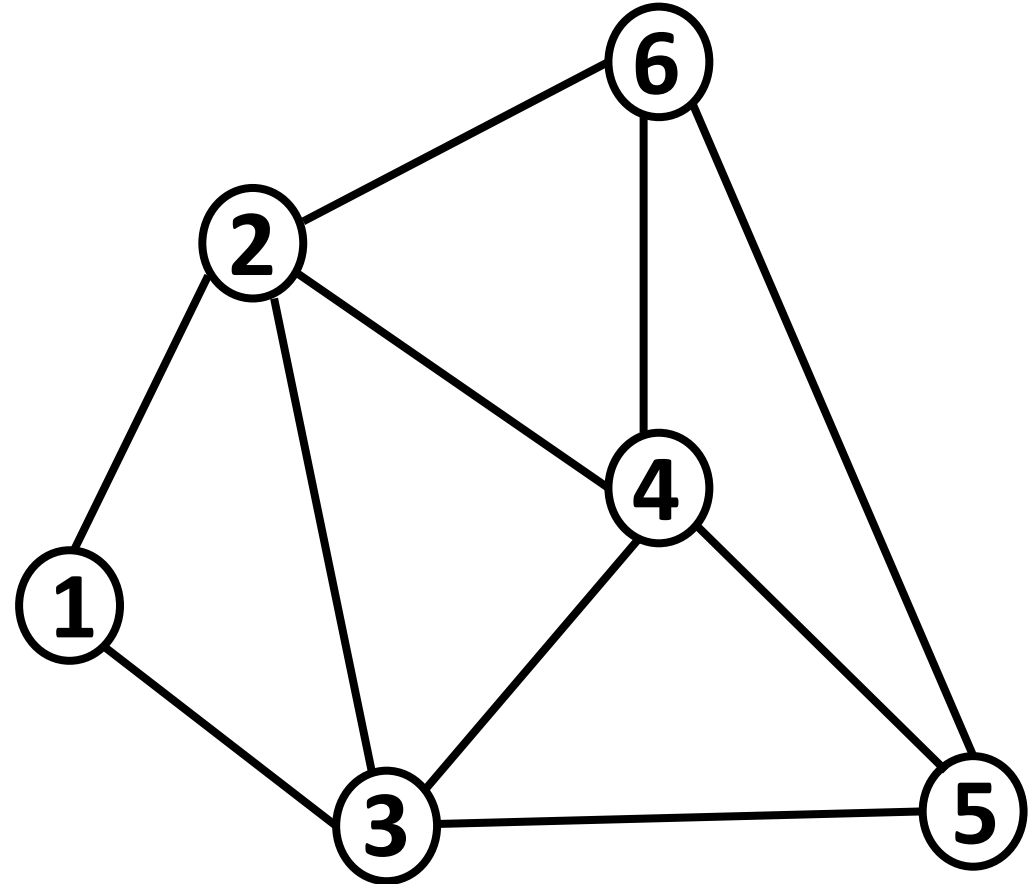
Part 3: Proofs of Infeasibility/Optimality

Part 4: Optimisation

Graph Colouring

$$x_i \neq x_j$$

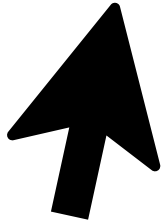
$$\begin{aligned}x_1 &= \{ \text{blue circle} \quad \text{orange circle} \quad \text{green circle} \} \\x_2 &= \{ \text{blue circle} \quad \text{orange circle} \quad \text{green circle} \} \\x_3 &= \{ \text{blue circle} \quad \text{orange circle} \quad \text{green circle} \} \\x_4 &= \{ \text{blue circle} \quad \text{orange circle} \quad \text{green circle} \} \\x_5 &= \{ \text{blue circle} \quad \text{orange circle} \quad \text{green circle} \} \\x_6 &= \{ \text{blue circle} \quad \text{orange circle} \quad \text{green circle} \}\end{aligned}$$



Graph Colouring

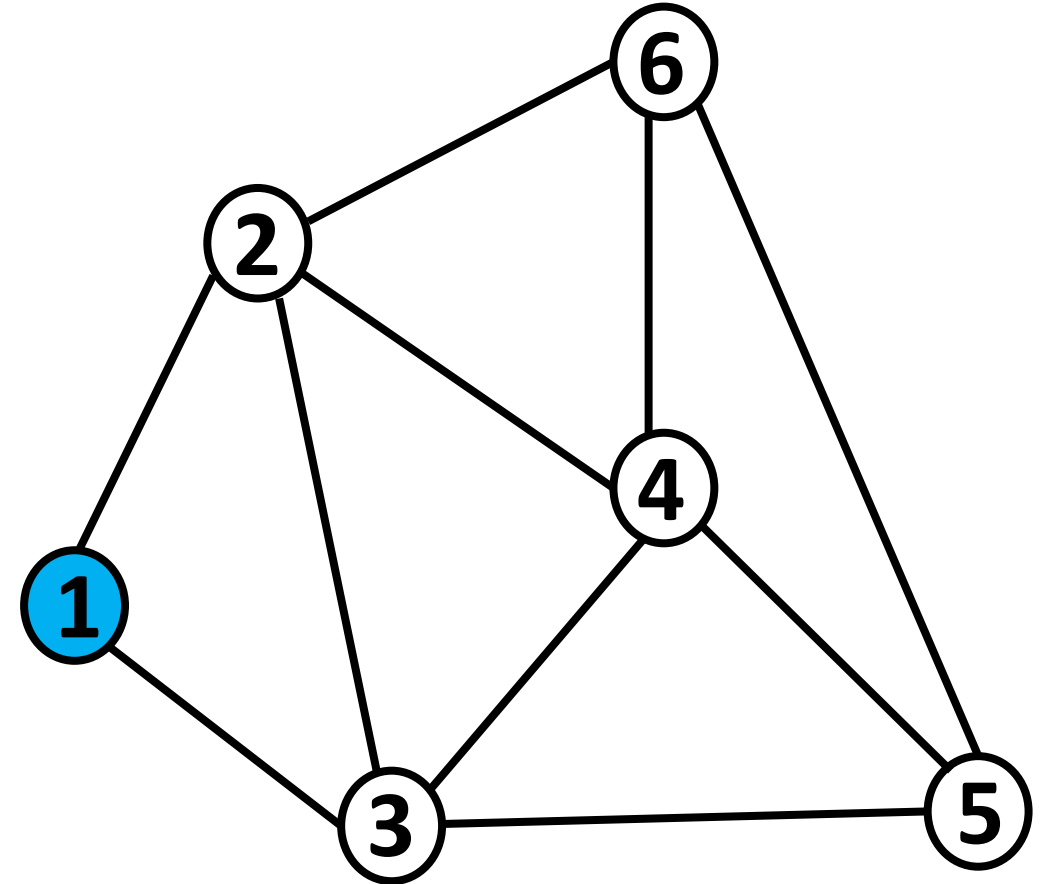
$$x_i \neq x_j$$

$$x_1 = \text{blue circle} @ 1$$



decision

$$\begin{aligned} x_1 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_2 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_3 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_4 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_5 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_6 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \end{aligned}$$

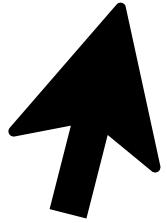


Graph Colouring

$$x_i \neq x_j$$

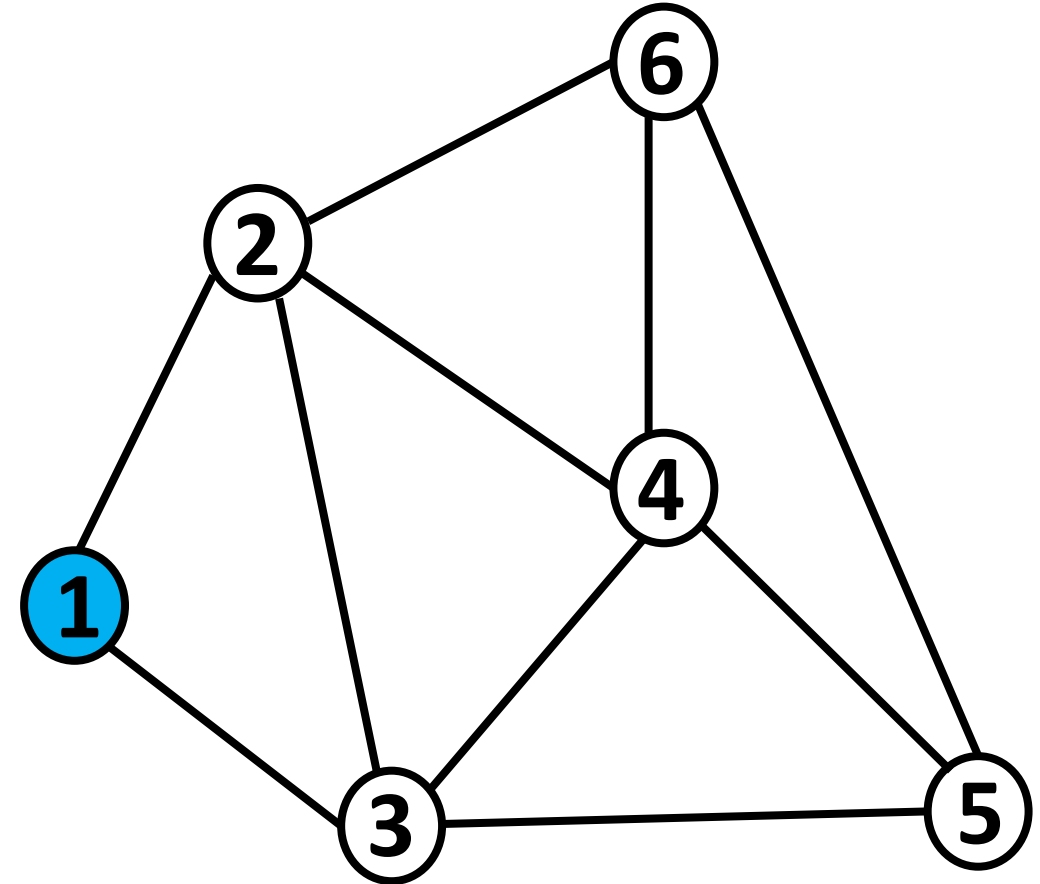
$$x_1 = \text{blue circle} @ 1$$

$x_2 \neq \text{blue circle} @ 1$
 $x_3 \neq \text{blue circle} @ 1$



propagations

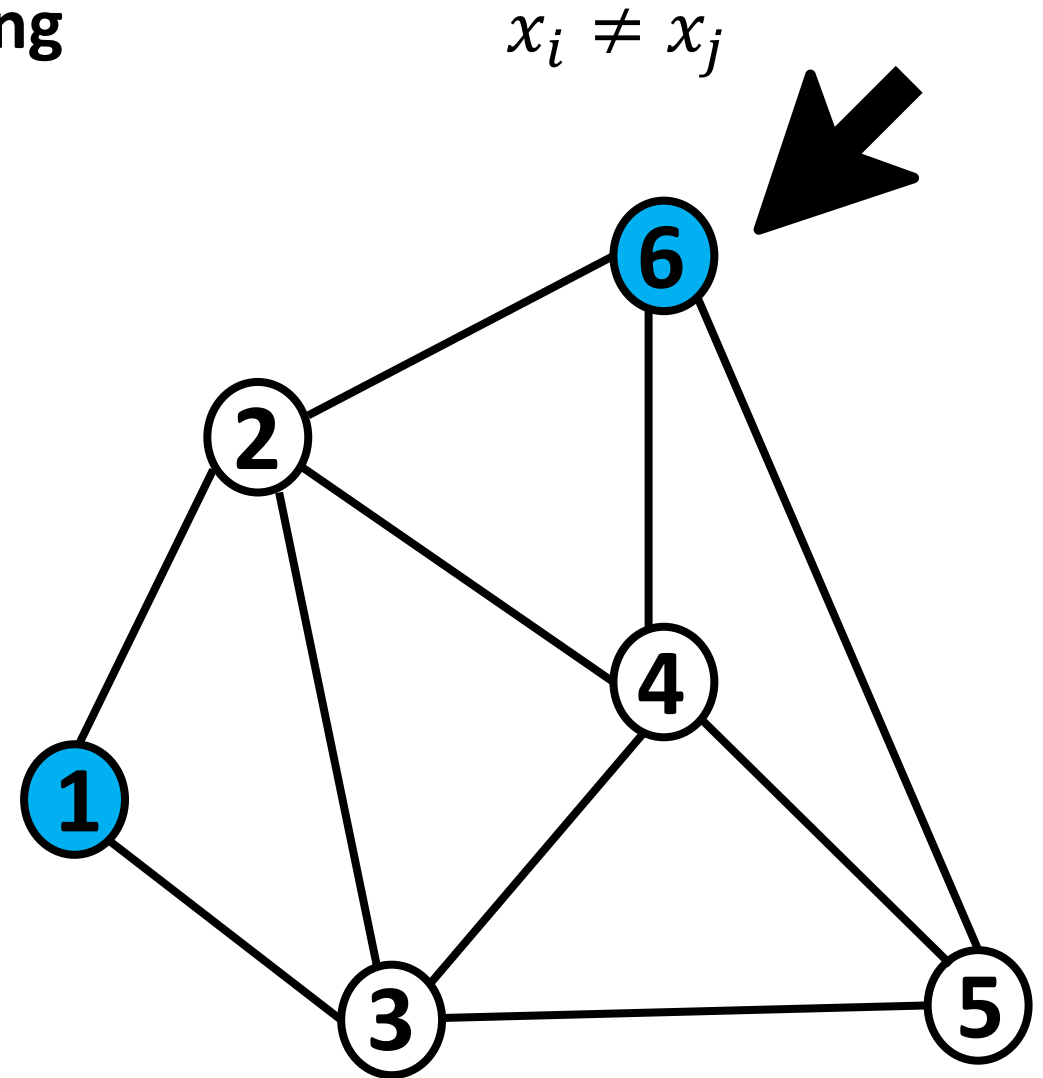
$x_1 = \{$				$\}$
$x_2 = \{$				$\}$
$x_3 = \{$				$\}$
$x_4 = \{$				$\}$
$x_5 = \{$				$\}$
$x_6 = \{$				$\}$



Graph Colouring

$x_1 = \text{blue circle @ 1}$
 $x_2 \neq \text{blue circle @ 1}$
 $x_3 \neq \text{blue circle @ 1}$
 $x_6 = \text{blue circle @ 2}$

$x_1 = \{$				$\}$
$x_2 = \{$				$\}$
$x_3 = \{$				$\}$
$x_4 = \{$				$\}$
$x_5 = \{$				$\}$
$x_6 = \{$				$\}$



Graph Colouring

$$x_i \neq x_j$$

$x_1 = \text{blue circle @ 1}$

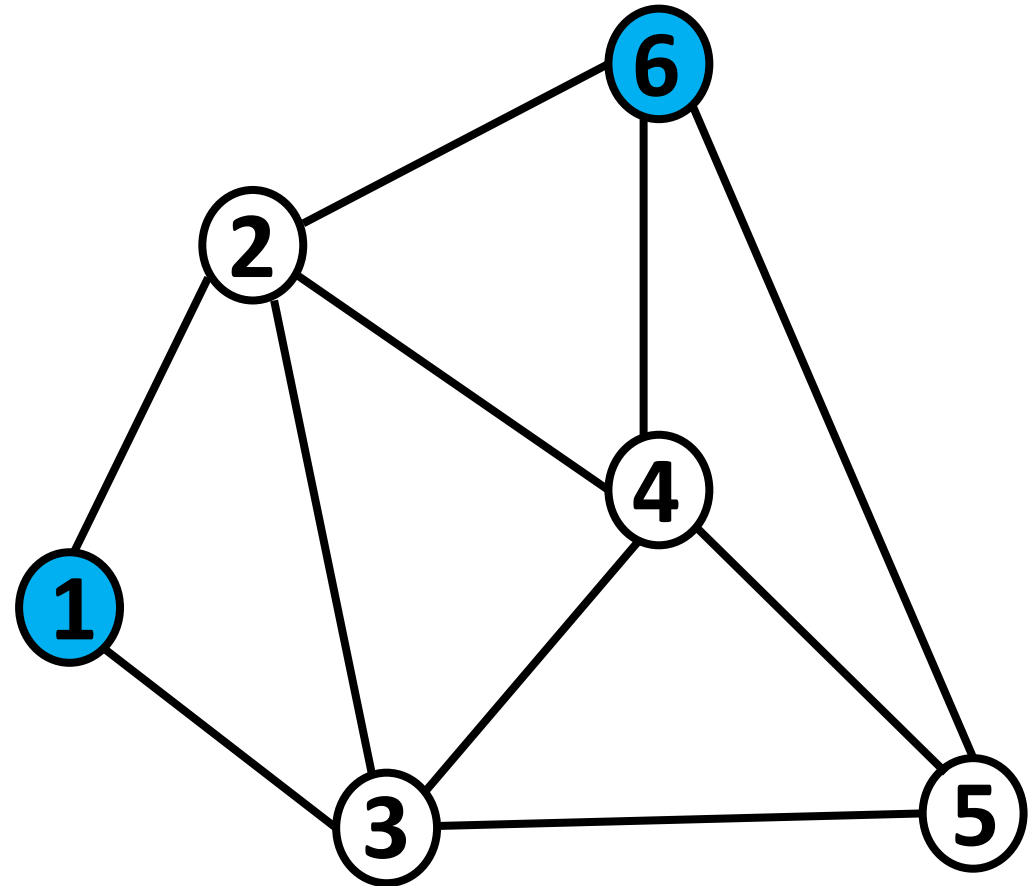
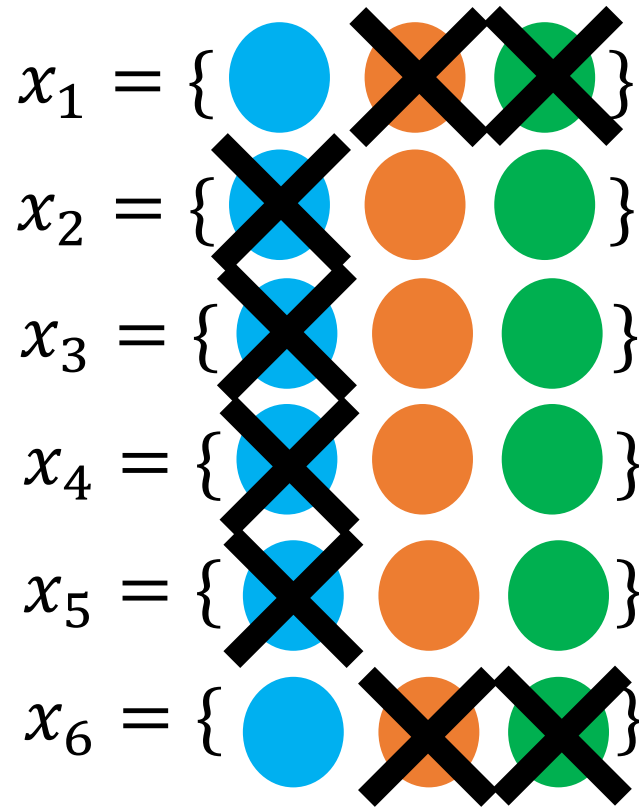
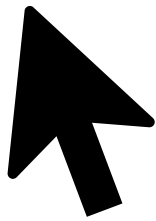
$x_2 \neq \text{blue circle @ 1}$

$x_3 \neq \text{blue circle @ 1}$

$x_6 = \text{blue circle @ 2}$

$x_4 \neq \text{blue circle @ 2}$

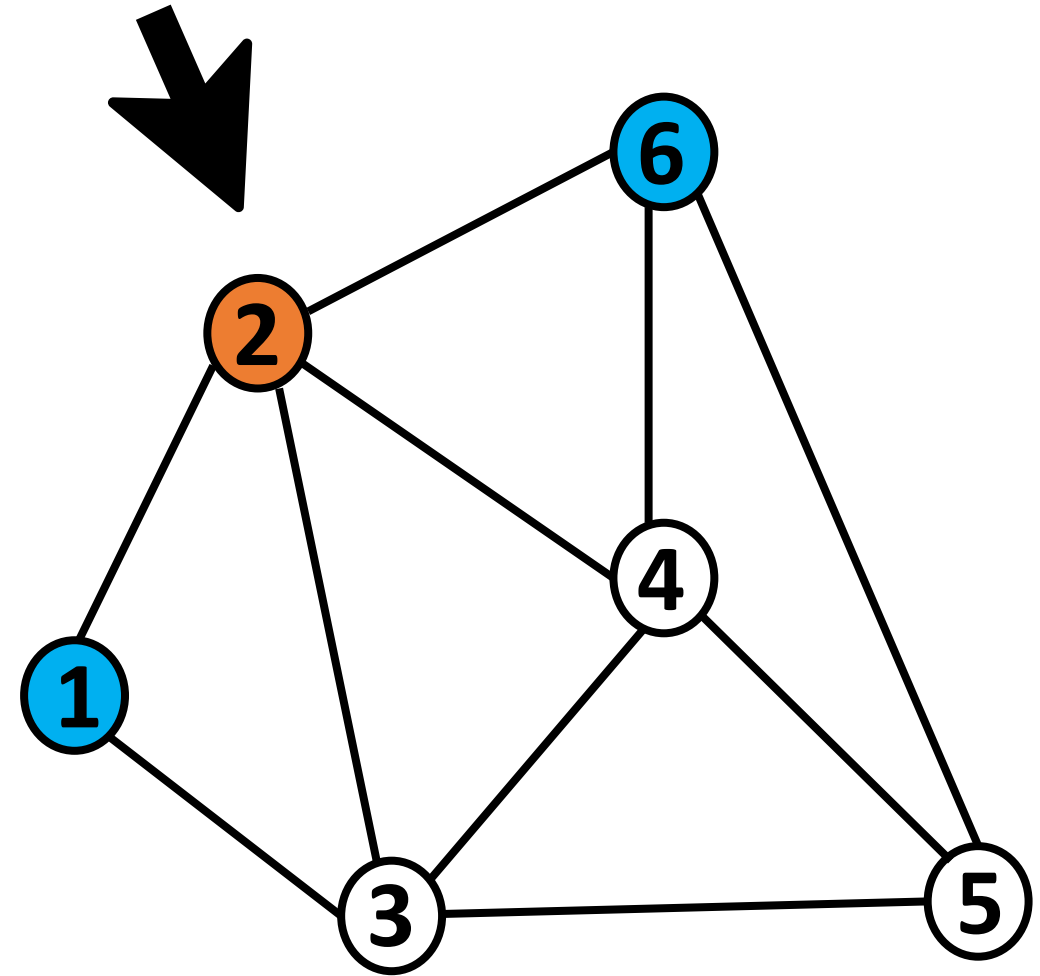
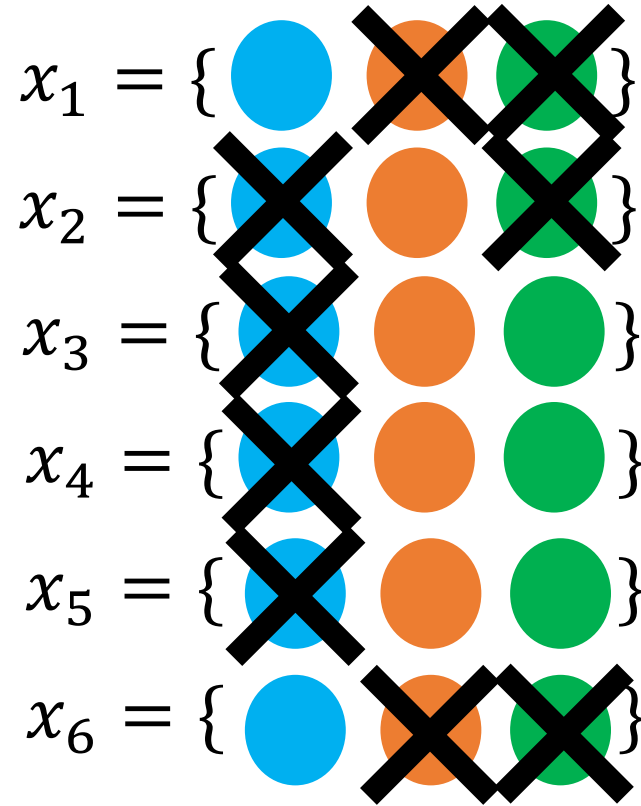
$x_5 \neq \text{blue circle @ 2}$



Graph Colouring

$$x_i \neq x_j$$

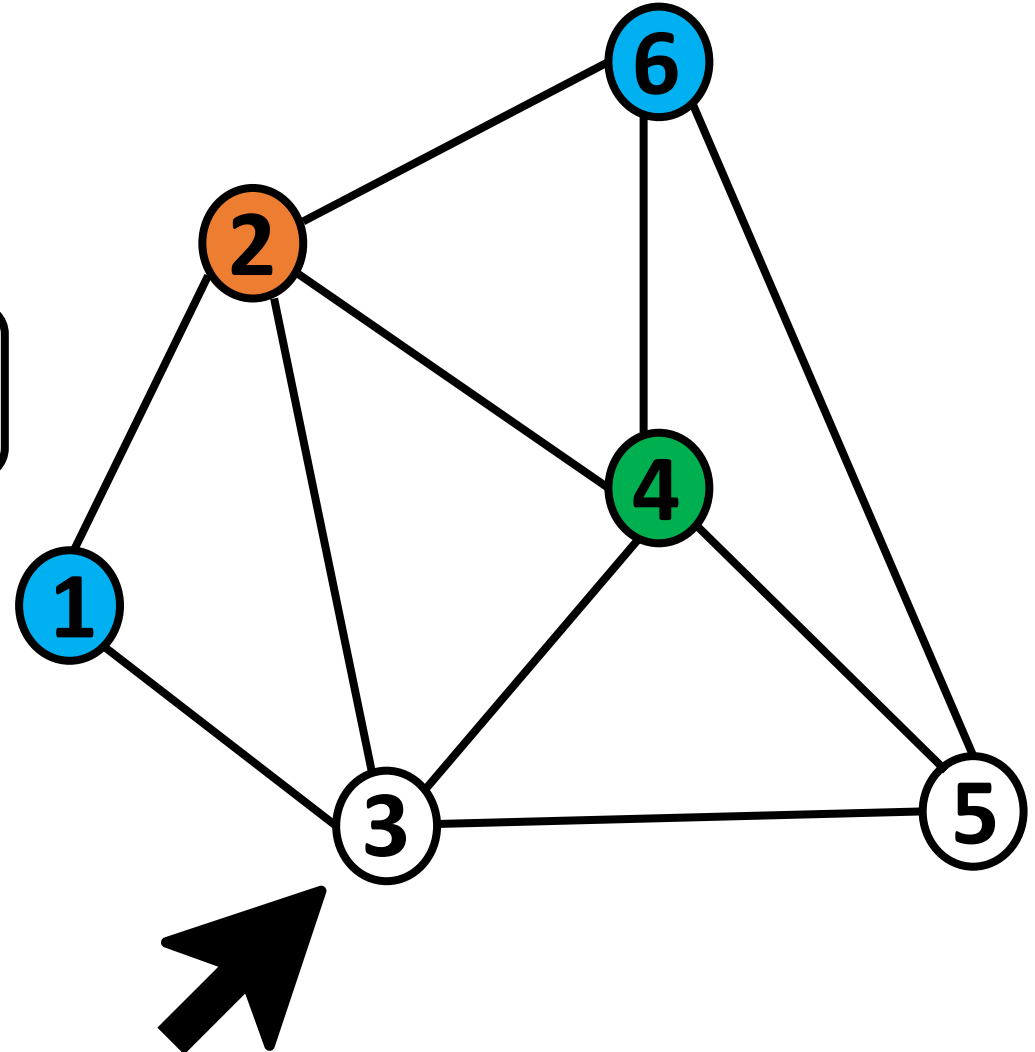
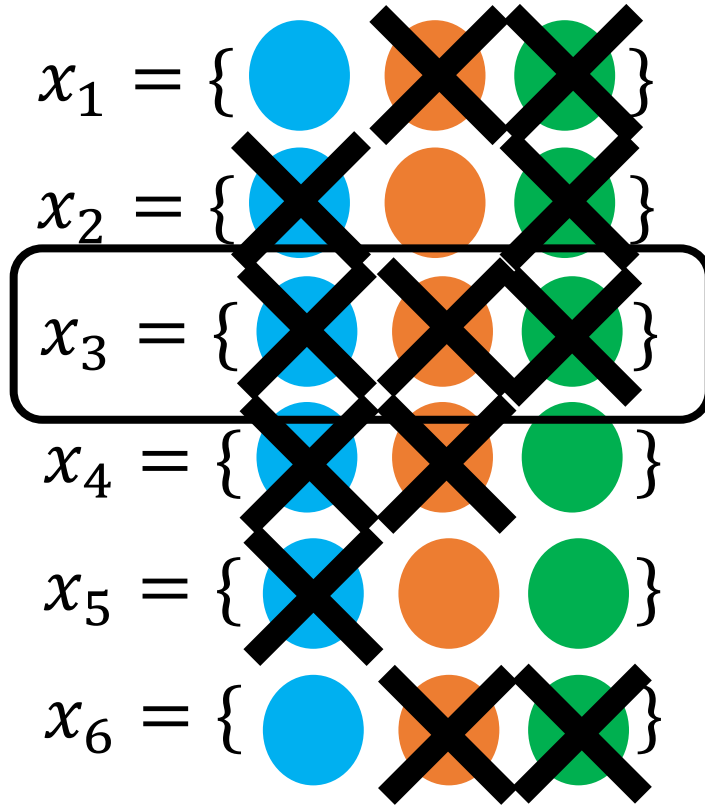
$x_1 = \text{blue circle @ 1}$
 $x_2 \neq \text{blue circle @ 1}$
 $x_3 \neq \text{blue circle @ 1}$
 $x_6 = \text{blue circle @ 2}$
 $x_4 \neq \text{blue circle @ 2}$
 $x_5 \neq \text{blue circle @ 2}$
 $x_2 = \text{orange circle @ 3}$



Graph Colouring

$$x_i \neq x_j$$

$x_1 =$  @ 1
 $x_2 \neq$  @ 1
 $x_3 \neq$  @ 1
 $x_6 =$  @ 2
 $x_4 \neq$  @ 2
 $x_5 \neq$  @ 2
 $x_2 =$  @ 3
 $x_4 \neq$  @ 3
 $x_3 \neq$  @ 3

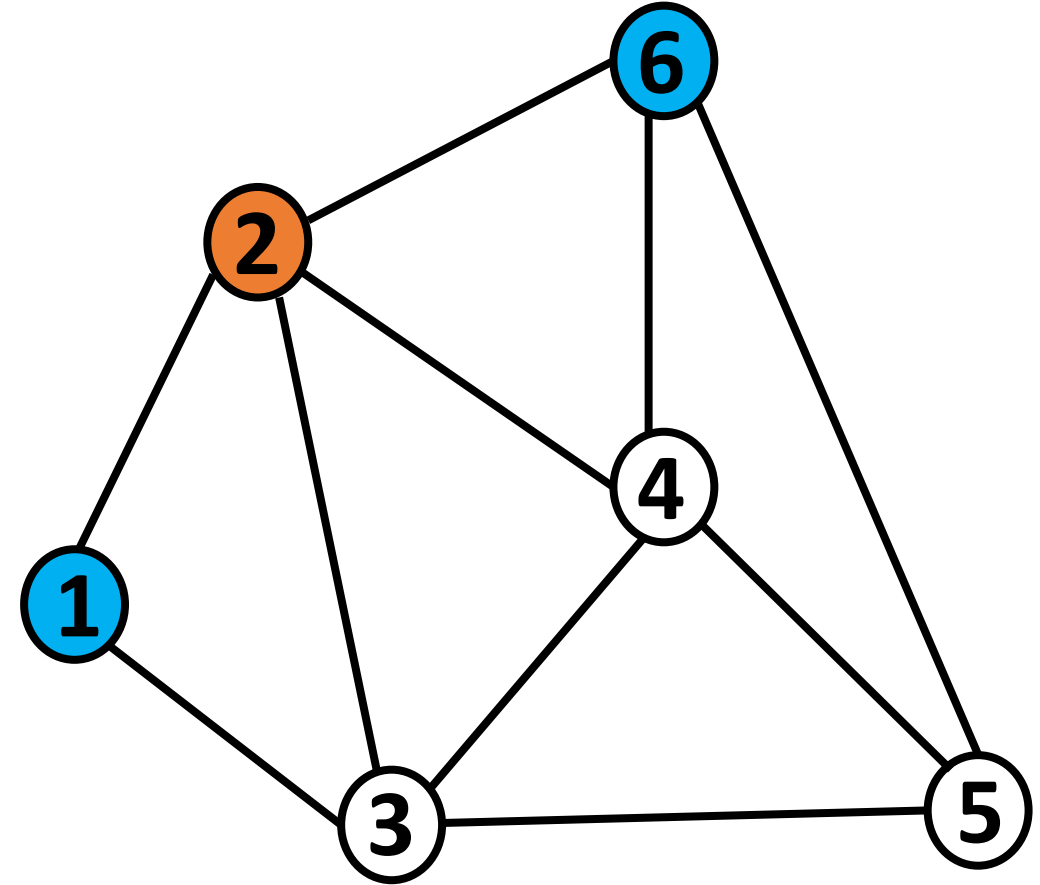
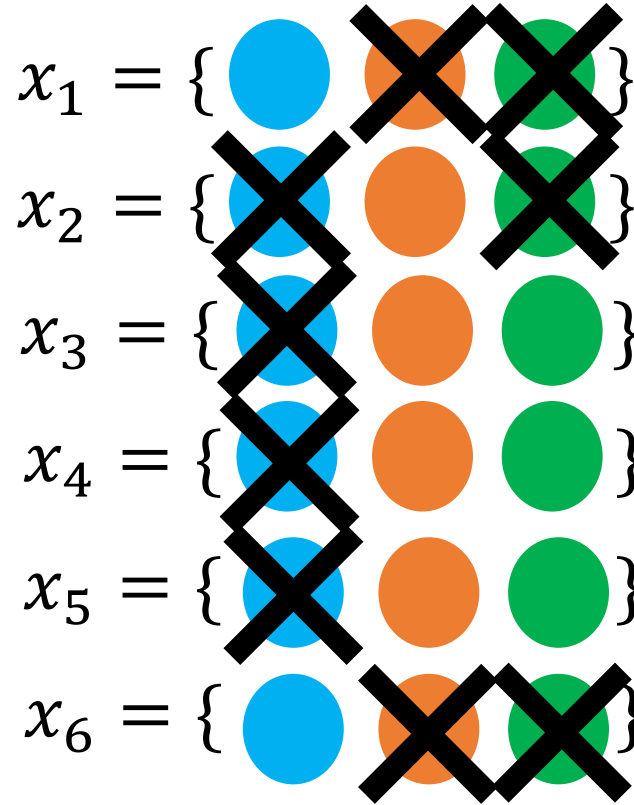


No colours left, conflict!

Graph Colouring

$$x_i \neq x_j$$

$x_1 = \text{blue} @ 1$
 $x_2 \neq \text{blue} @ 1$
 $x_3 \neq \text{blue} @ 1$
 $x_6 = \text{blue} @ 2$
 $x_4 \neq \text{blue} @ 2$
 $x_5 \neq \text{blue} @ 2$
 $x_2 = \text{orange} @ 3$

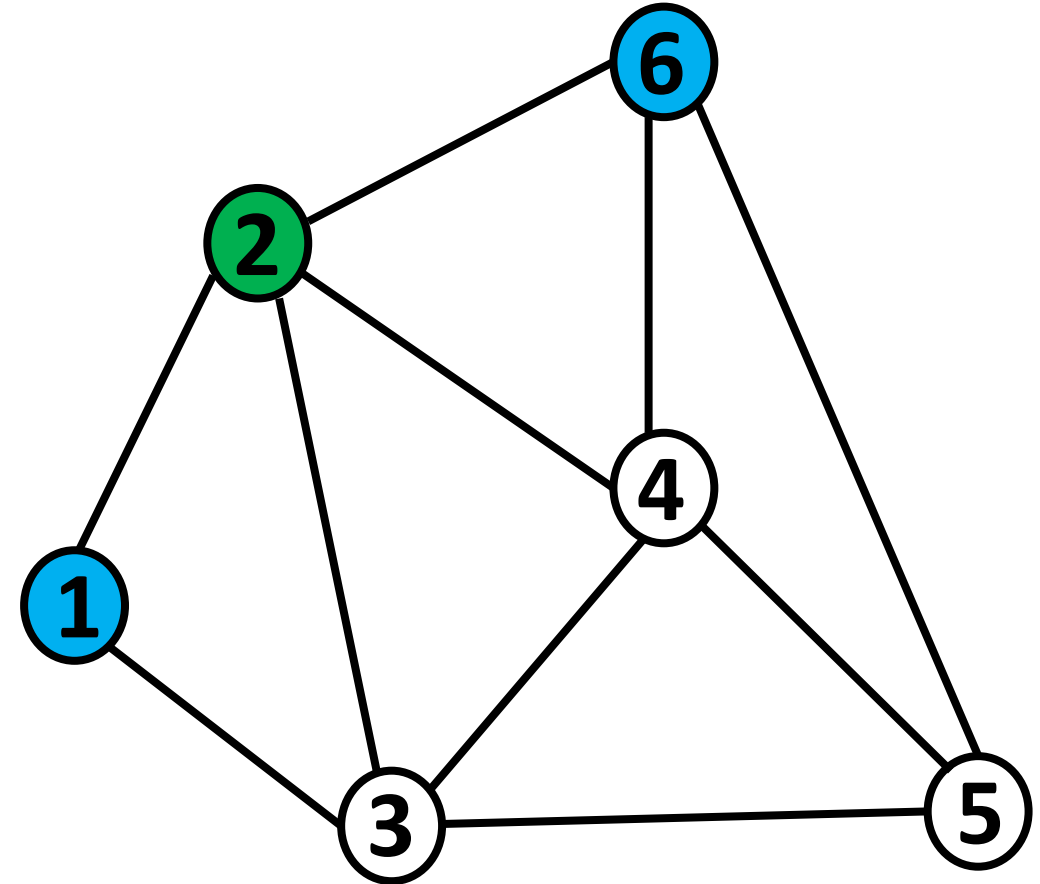
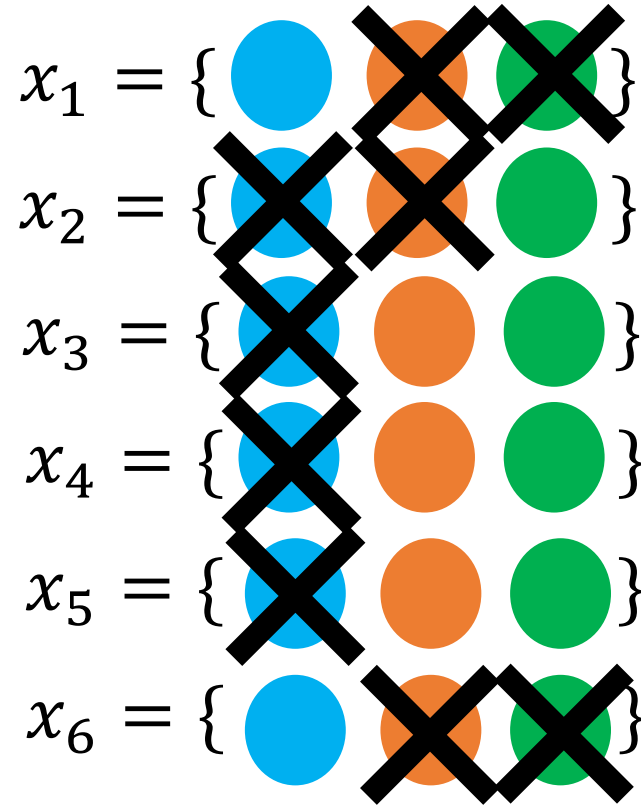


Go back to last decision

Graph Colouring

$$x_i \neq x_j$$

$x_1 = \text{blue} @ 1$
 $x_2 \neq \text{blue} @ 1$
 $x_3 \neq \text{blue} @ 1$
 $x_6 = \text{blue} @ 2$
 $x_4 \neq \text{blue} @ 2$
 $x_5 \neq \text{blue} @ 2$
 $x_2 \neq \text{orange} @ 2$

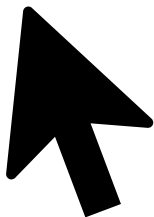


We now know that orange was in fact not possible

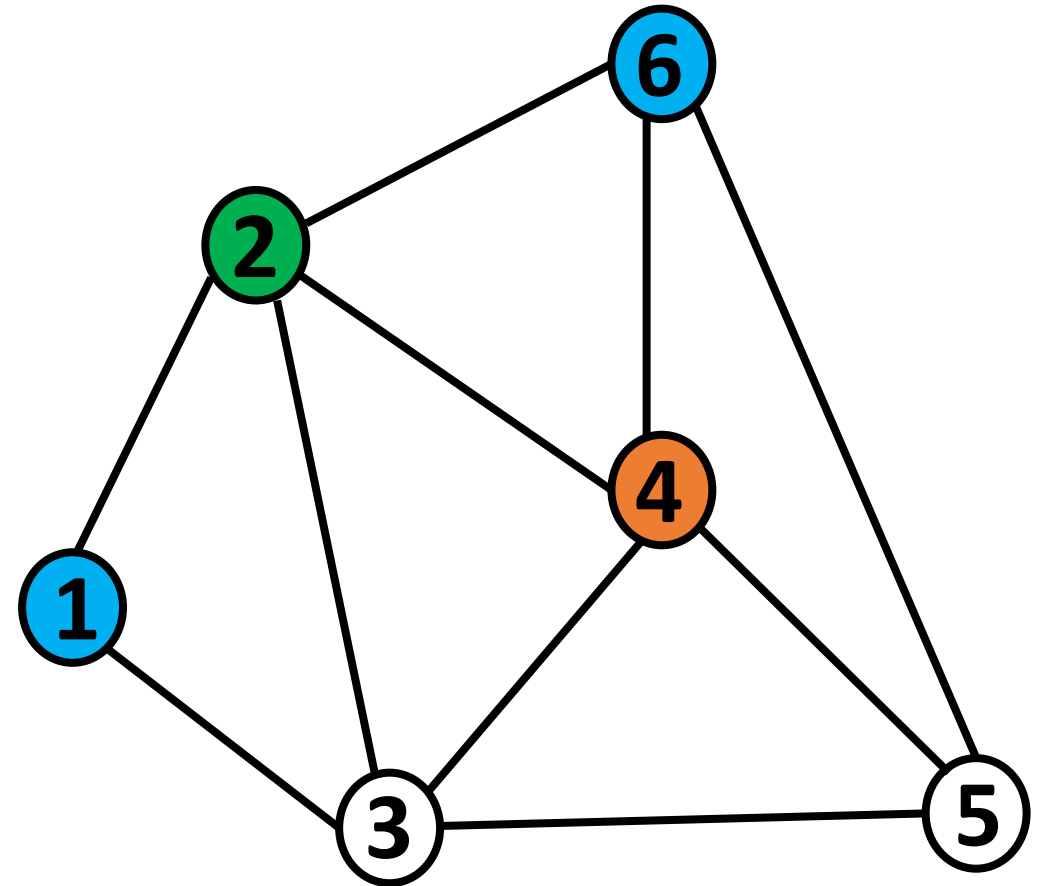
Graph Colouring

$$x_i \neq x_j$$

$x_1 =$  @ 1
 $x_2 \neq$  @ 1
 $x_3 \neq$  @ 1
 $x_6 =$  @ 2
 $x_4 \neq$  @ 2
 $x_5 \neq$  @ 2
 $x_2 \neq$  @ 2
 $x_4 \neq$  @ 2



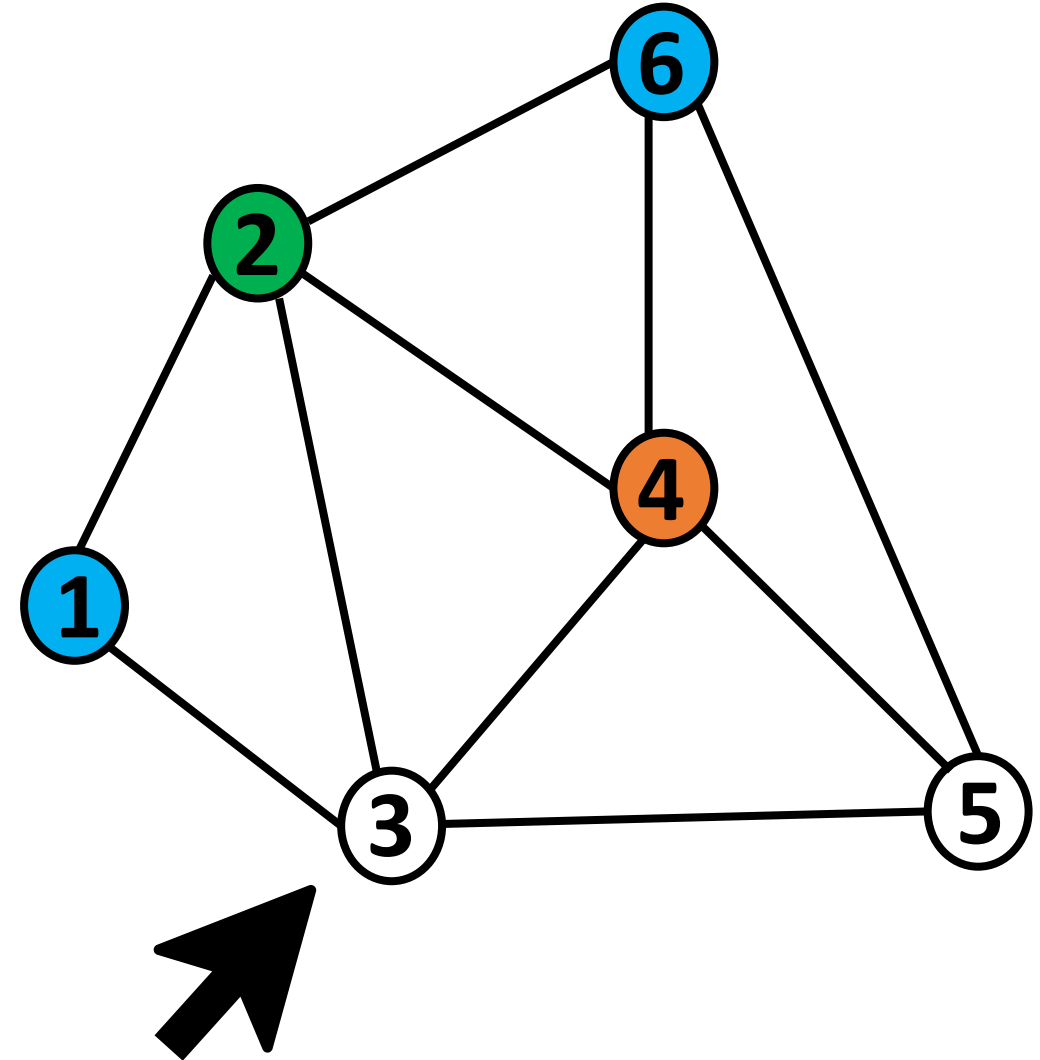
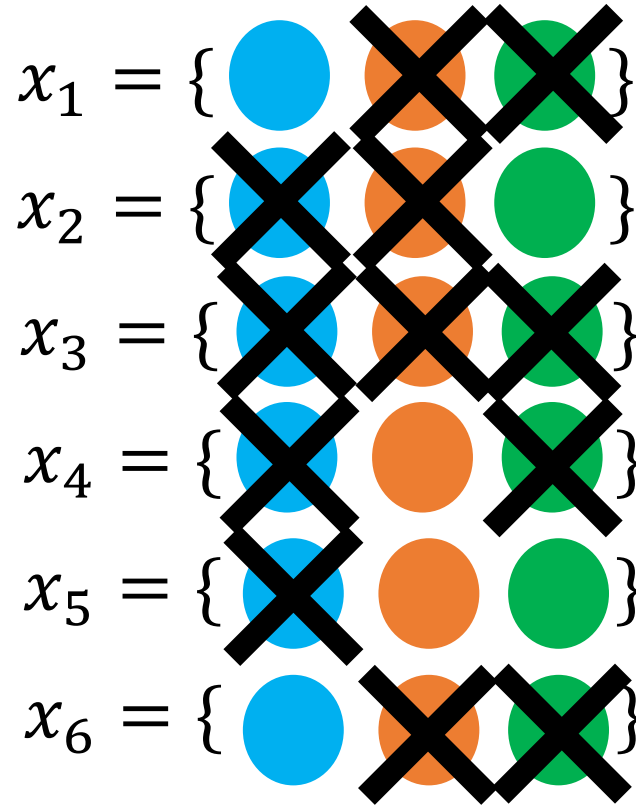
$x_1 = \{$    $\}$
 $x_2 = \{$    $\}$
 $x_3 = \{$    $\}$
 $x_4 = \{$    $\}$
 $x_5 = \{$    $\}$
 $x_6 = \{$    $\}$



Graph Colouring

$$x_i \neq x_j$$

$x_1 =$  @ 1
 $x_2 \neq$  @ 1
 $x_3 \neq$  @ 1
 $x_6 =$  @ 2
 $x_4 \neq$  @ 2
 $x_5 \neq$  @ 2
 $x_2 \neq$  @ 2
 $x_4 \neq$  @ 2
 $x_3 \neq$  @ 2
 $x_3 \neq$  @ 2



No colours left, conflict!

Graph Colouring

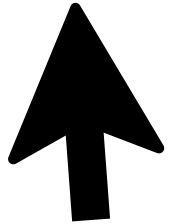
$$x_i \neq x_j$$

$$x_1 = \text{blue circle} @ 1$$

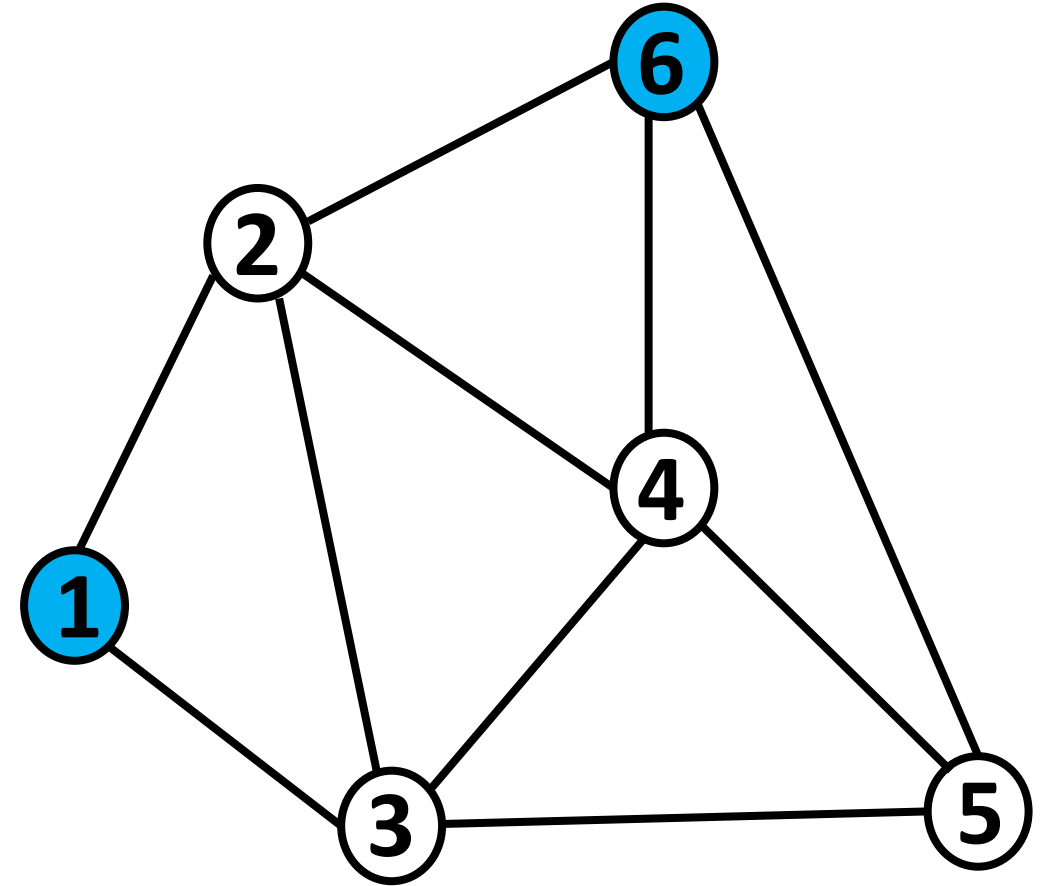
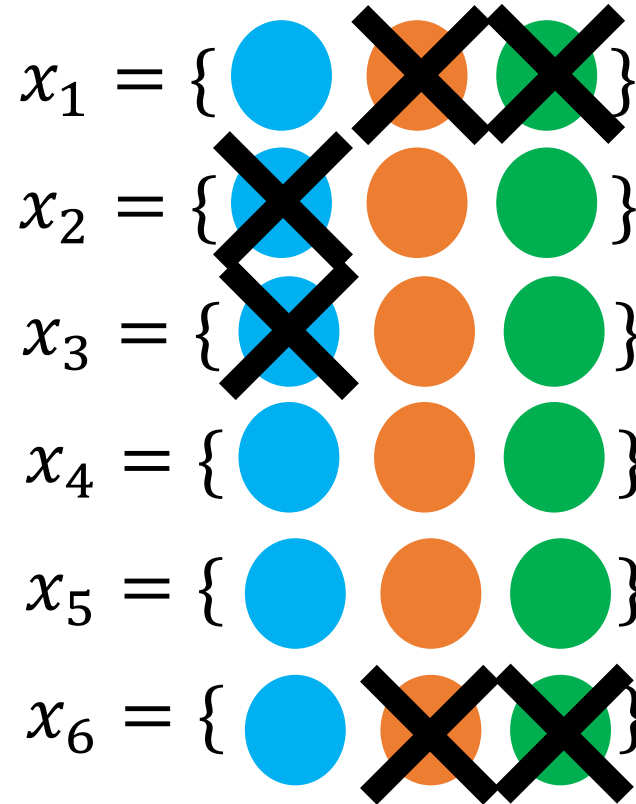
$$x_2 \neq \text{blue circle} @ 1$$

$$x_3 \neq \text{blue circle} @ 1$$

$$x_6 = \text{blue circle} @ 2$$



Go back to
last decision



Graph Colouring

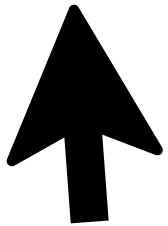
$$x_i \neq x_j$$

$$x_1 = \text{blue} @ 1$$

$$x_2 \neq \text{blue} @ 1$$

$$x_3 \neq \text{blue} @ 1$$

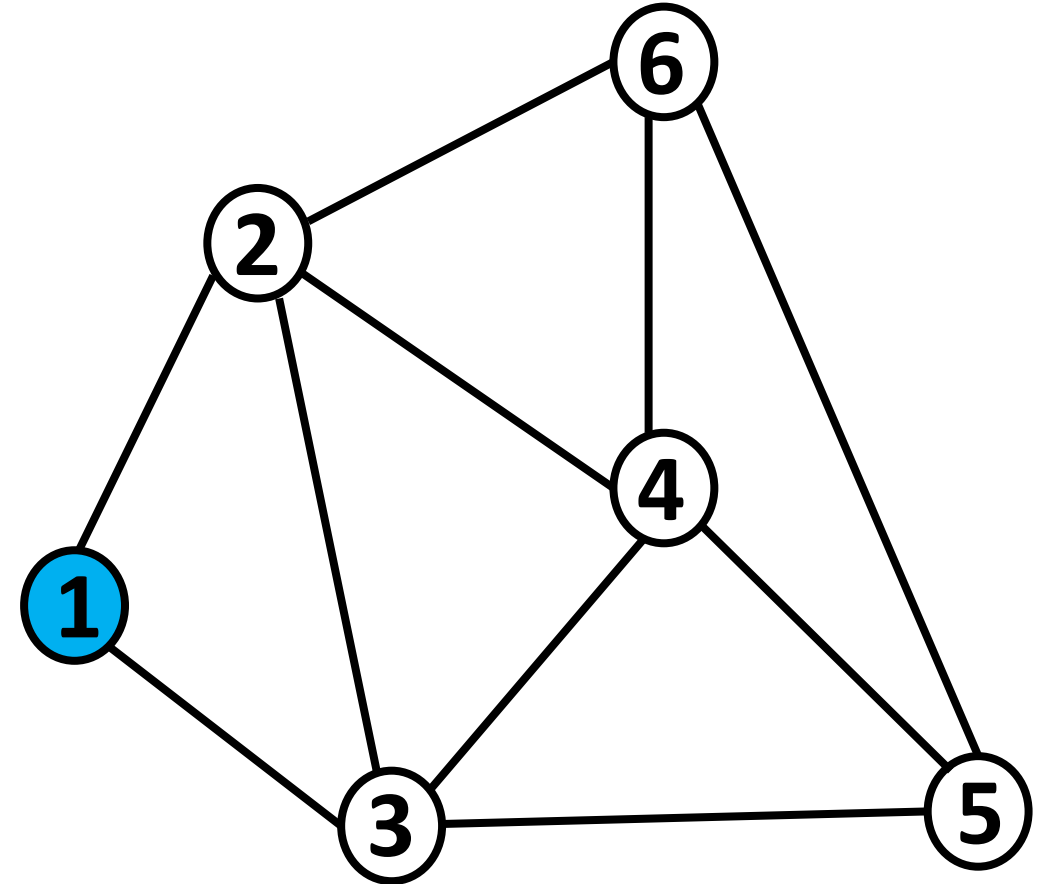
$$x_6 \neq \text{blue} @ 1$$



We now know
that blue
was in fact
not possible

$$\begin{aligned} x_1 &= \{ \text{blue}, \text{orange}, \text{green} \} \\ x_2 &= \{ \text{blue}, \text{orange}, \text{green} \} \\ x_3 &= \{ \text{blue}, \text{orange}, \text{green} \} \\ x_4 &= \{ \text{blue}, \text{orange}, \text{green} \} \\ x_5 &= \{ \text{blue}, \text{orange}, \text{green} \} \\ x_6 &= \{ \text{blue}, \text{orange}, \text{green} \} \end{aligned}$$

The above set of equations shows a 3x6 grid of colored circles (blue, orange, green) with black 'X' marks over the blue circles in the first, second, third, and sixth rows, indicating that blue is not a valid color for those nodes.



Graph Colouring

$$x_i \neq x_j$$

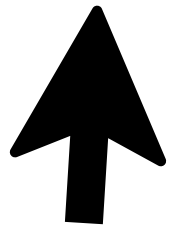
$$x_1 = \text{blue circle} @ 1$$

$$x_2 \neq \text{blue circle} @ 1$$

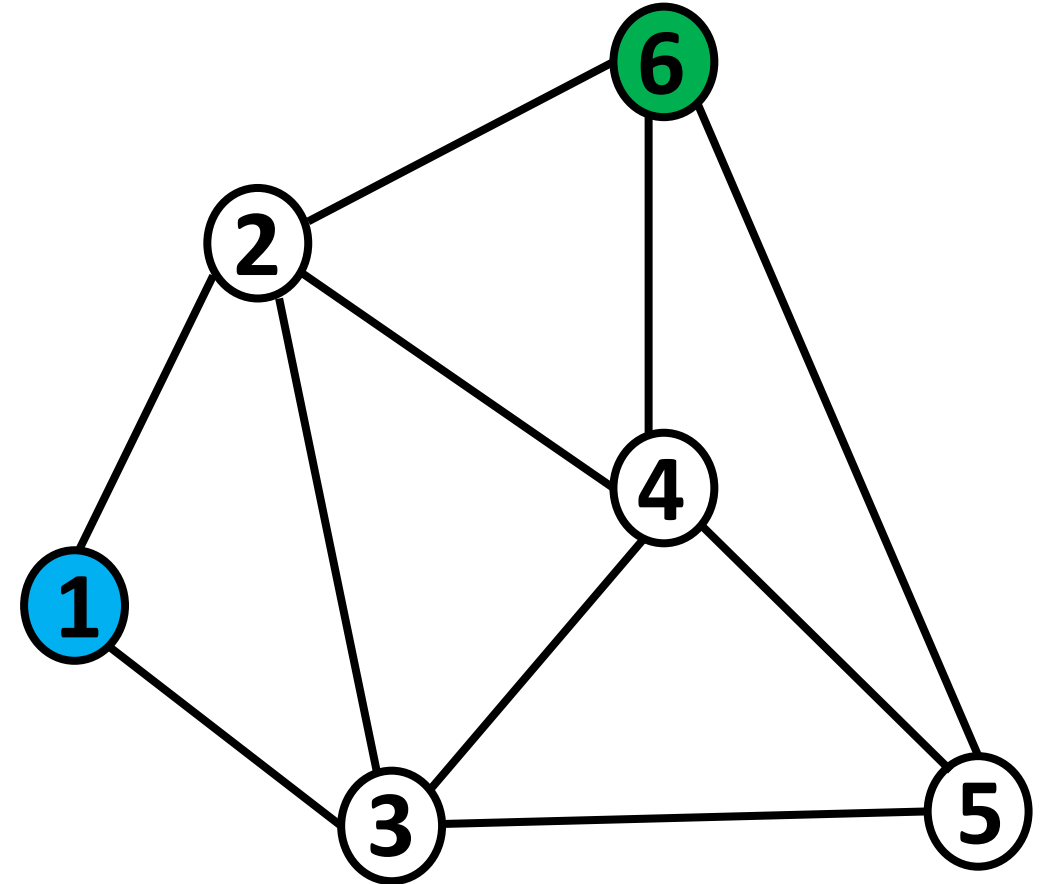
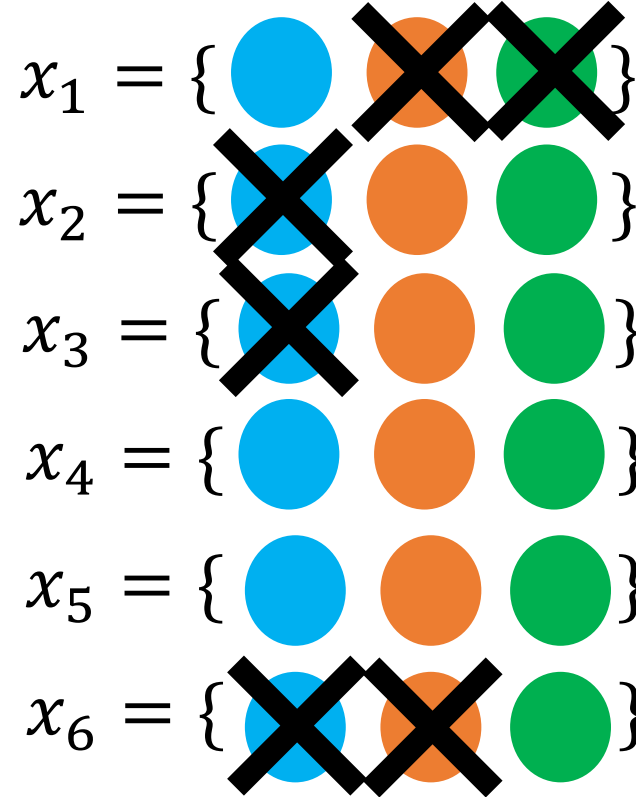
$$x_3 \neq \text{blue circle} @ 1$$

$$x_6 \neq \text{blue circle} @ 1$$

$$x_6 = \text{green circle} @ 2$$



new decision



Graph Colouring

$$x_i \neq x_j$$

$$x_1 = \text{blue circle} @ 1$$

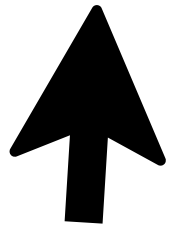
$$x_2 \neq \text{blue circle} @ 1$$

$$x_3 \neq \text{blue circle} @ 1$$

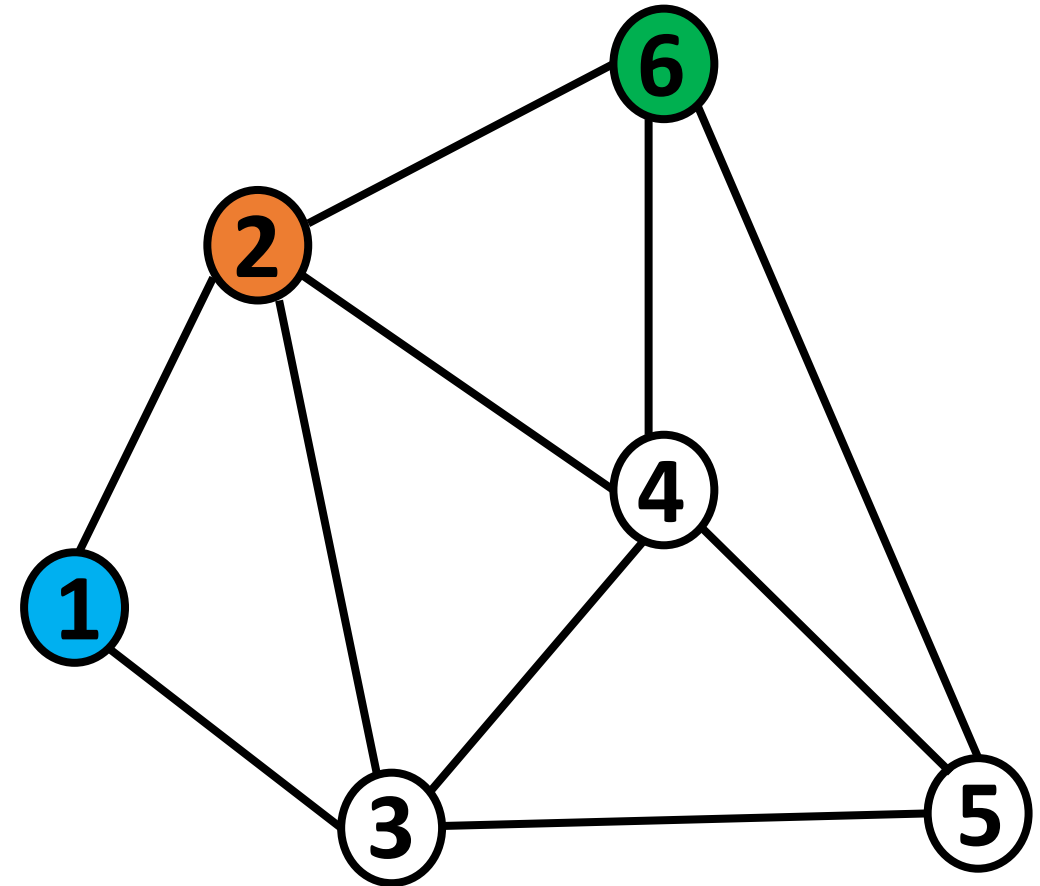
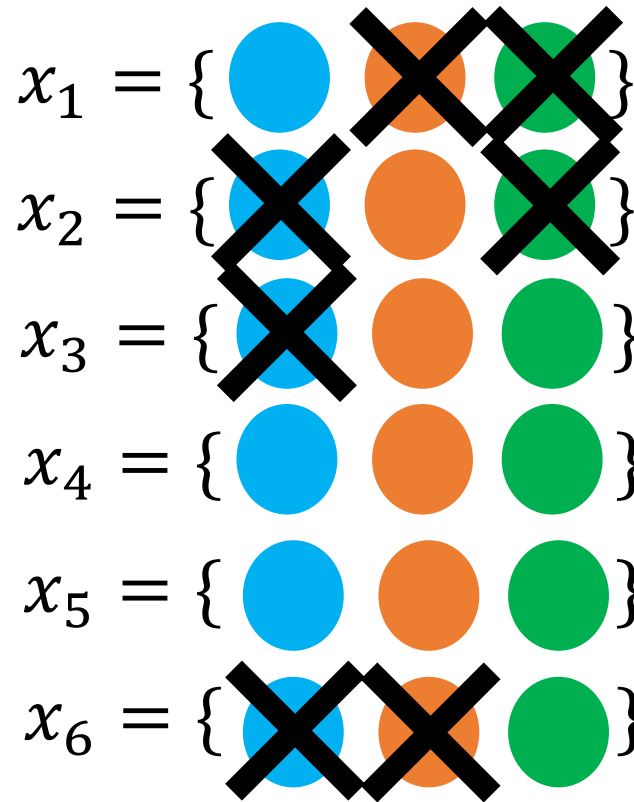
$$x_6 \neq \text{blue circle} @ 1$$

$$x_6 = \text{green circle} @ 2$$

$$x_2 \neq \text{green circle} @ 2$$



propagation



Graph Colouring

$$x_i \neq x_j$$

$$x_1 = \text{blue circle} @ 1$$

$$x_2 \neq \text{blue circle} @ 1$$

$$x_3 \neq \text{blue circle} @ 1$$

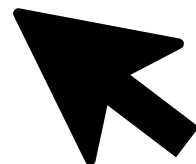
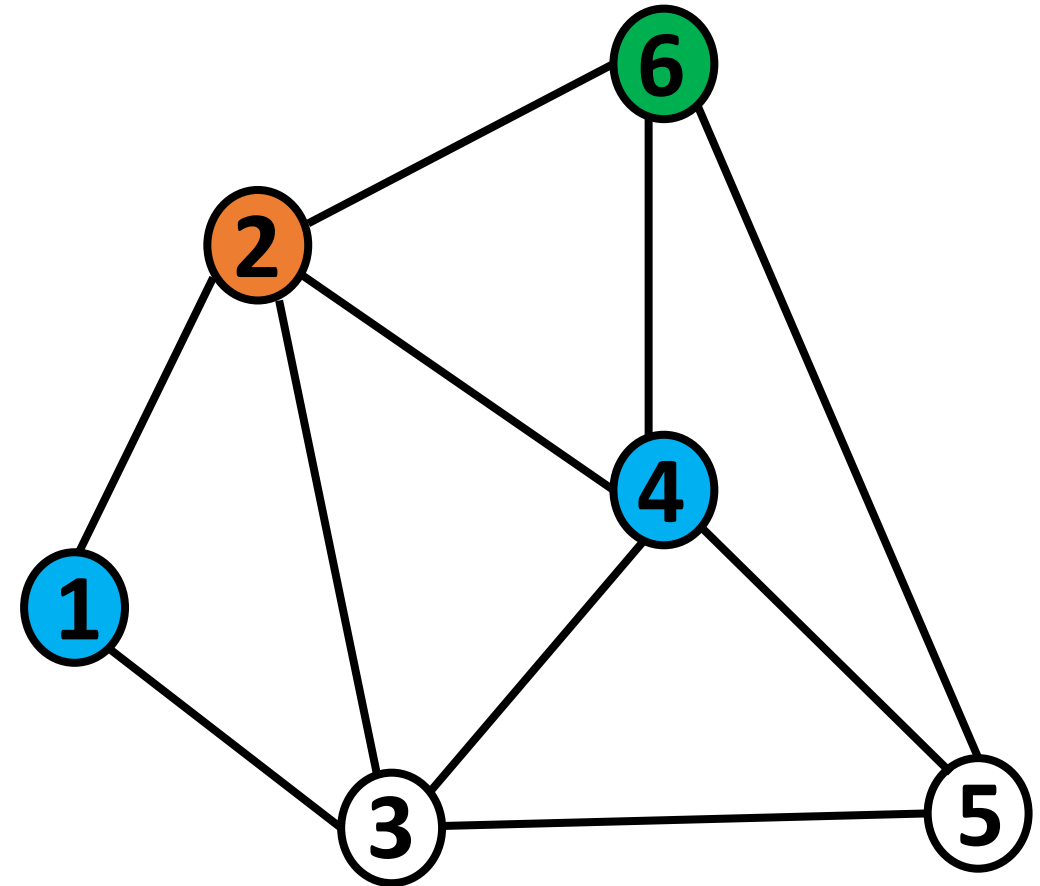
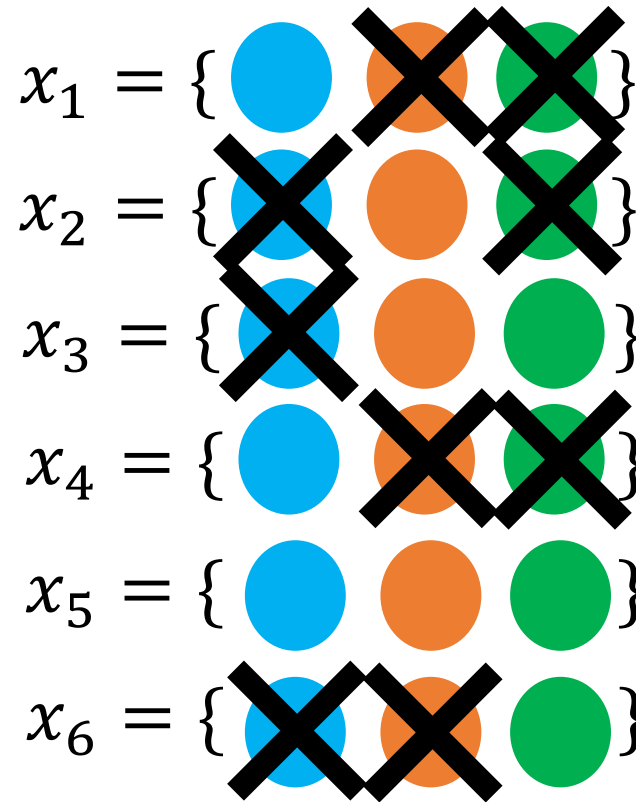
$$x_6 \neq \text{blue circle} @ 1$$

$$x_6 = \text{green circle} @ 2$$

$$x_2 \neq \text{green circle} @ 2$$

$$x_4 \neq \text{green circle} @ 2$$

$$x_4 \neq \text{orange circle} @ 2$$



propagation

Graph Colouring

$$x_i \neq x_j$$

$$x_1 = \text{blue circle} @ 1$$

$$x_2 \neq \text{blue circle} @ 1$$

$$x_3 \neq \text{blue circle} @ 1$$

$$x_6 \neq \text{blue circle} @ 1$$

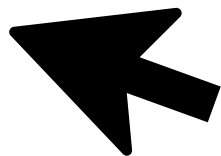
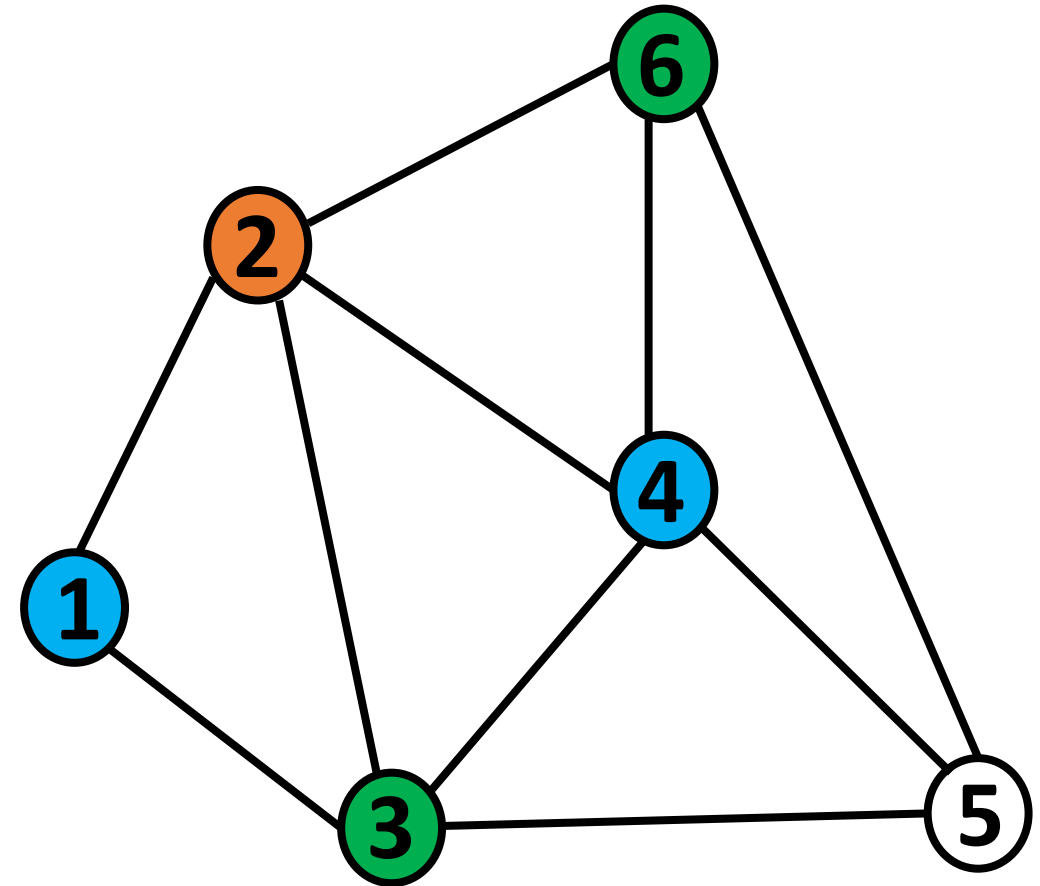
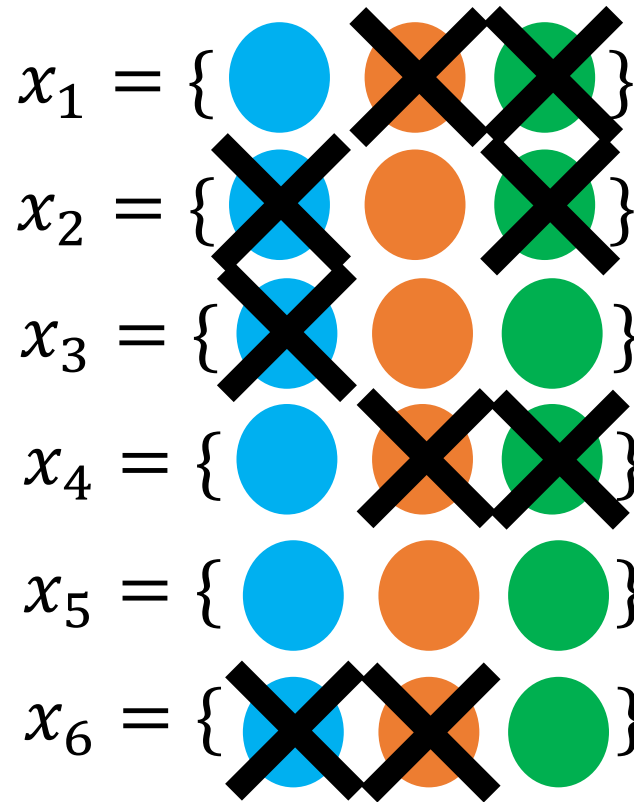
$$x_6 = \text{green circle} @ 2$$

$$x_2 \neq \text{green circle} @ 2$$

$$x_4 \neq \text{green circle} @ 2$$

$$x_4 \neq \text{orange circle} @ 2$$

$$x_3 \neq \text{orange circle} @ 2$$



Graph Colouring

$$x_i \neq x_j$$

$x_1 =$  @ 1

$x_2 \neq$  @ 1

$x_3 \neq$  @ 1

$x_6 \neq$  @ 1

$x_6 =$  @ 2

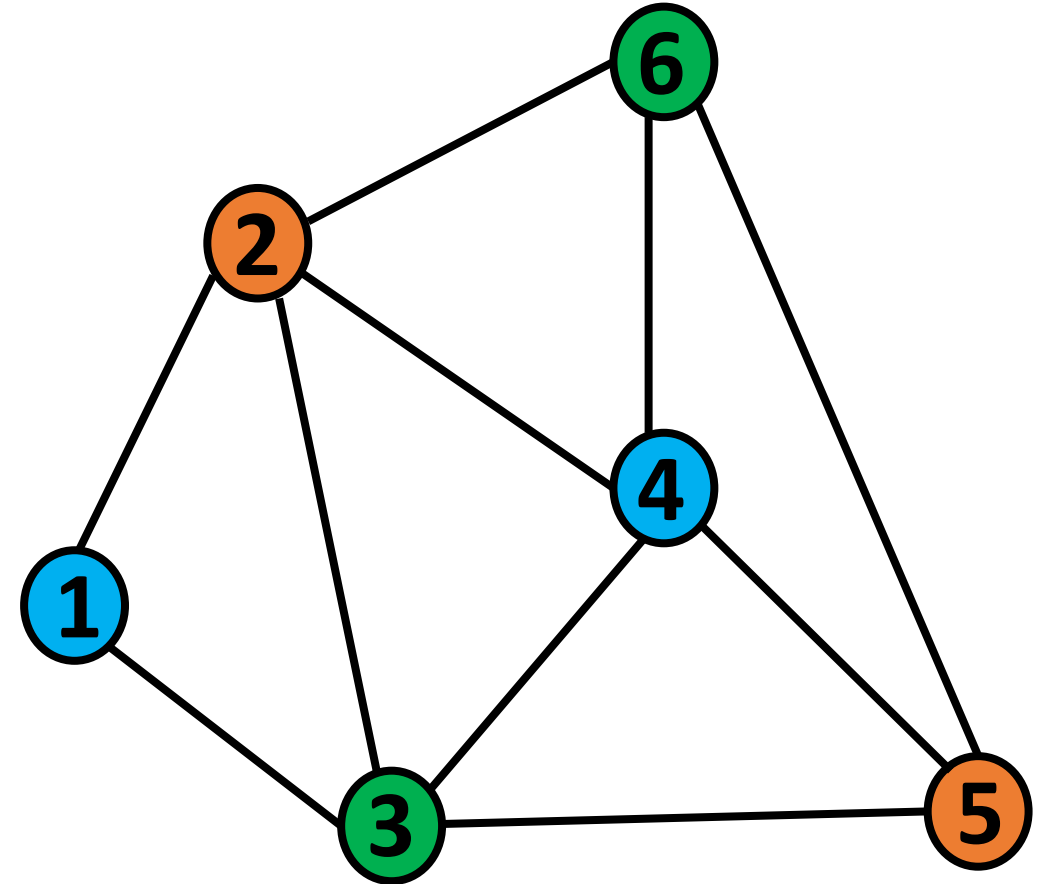
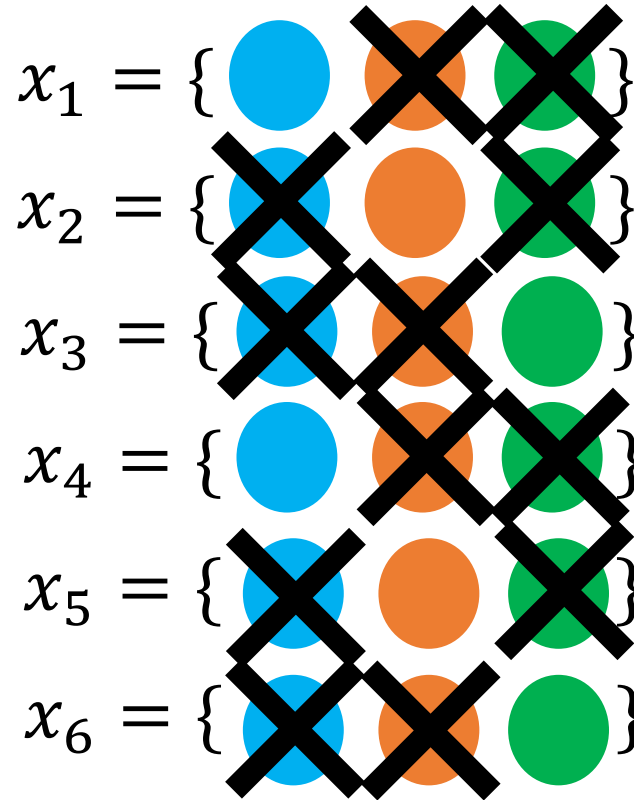
$x_2 \neq$  @ 2

$x_4 \neq$  @ 2

$x_4 \neq$  @ 2

$x_3 \neq$  @ 2

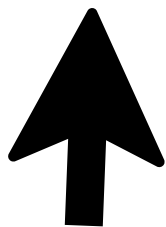
$x_5 =$  @ 2



Problem solved!



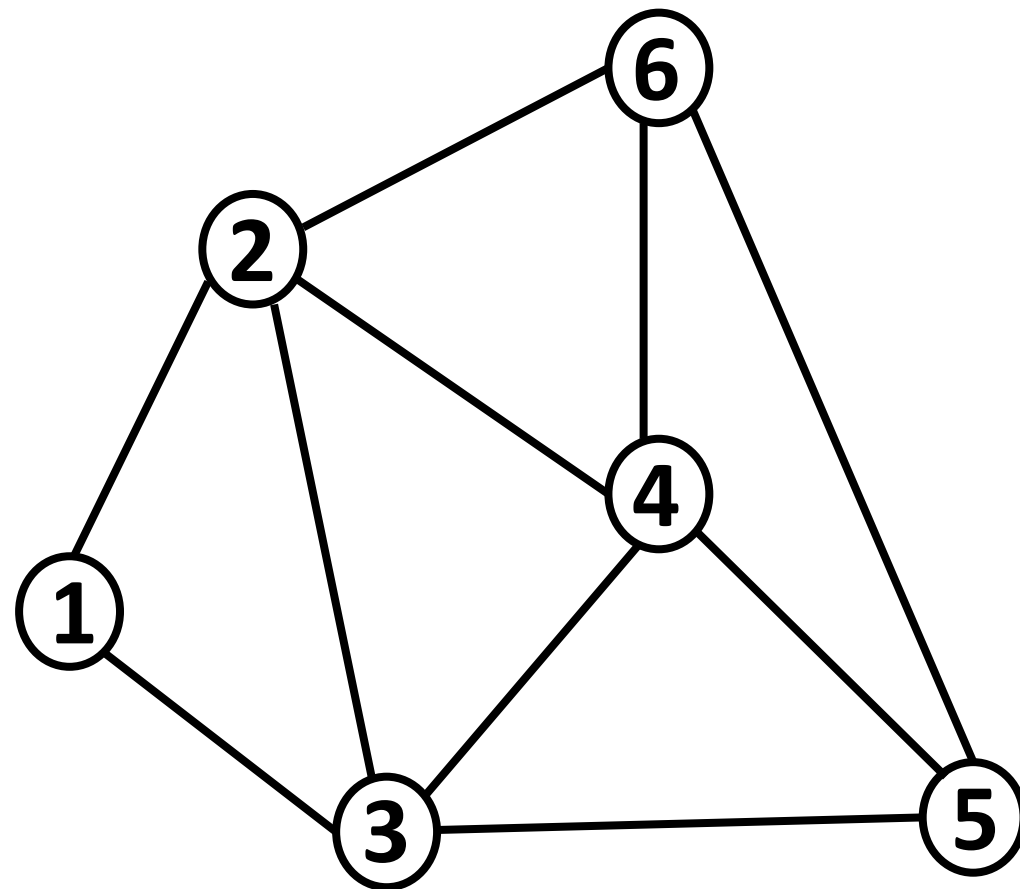
$$x_1 > x_2 > \dots > x_6$$



Graph Colouring

$$x_i \neq x_j$$

$$\begin{aligned} x_1 &= \{ \text{blue circle} \text{ , orange circle} \text{ , green circle} \} \\ x_2 &= \{ \text{blue circle} \text{ , orange circle} \text{ , green circle} \} \\ x_3 &= \{ \text{blue circle} \text{ , orange circle} \text{ , green circle} \} \\ x_4 &= \{ \text{blue circle} \text{ , orange circle} \text{ , green circle} \} \\ x_5 &= \{ \text{blue circle} \text{ , orange circle} \text{ , green circle} \} \\ x_6 &= \{ \text{blue circle} \text{ , orange circle} \text{ , green circle} \} \end{aligned}$$



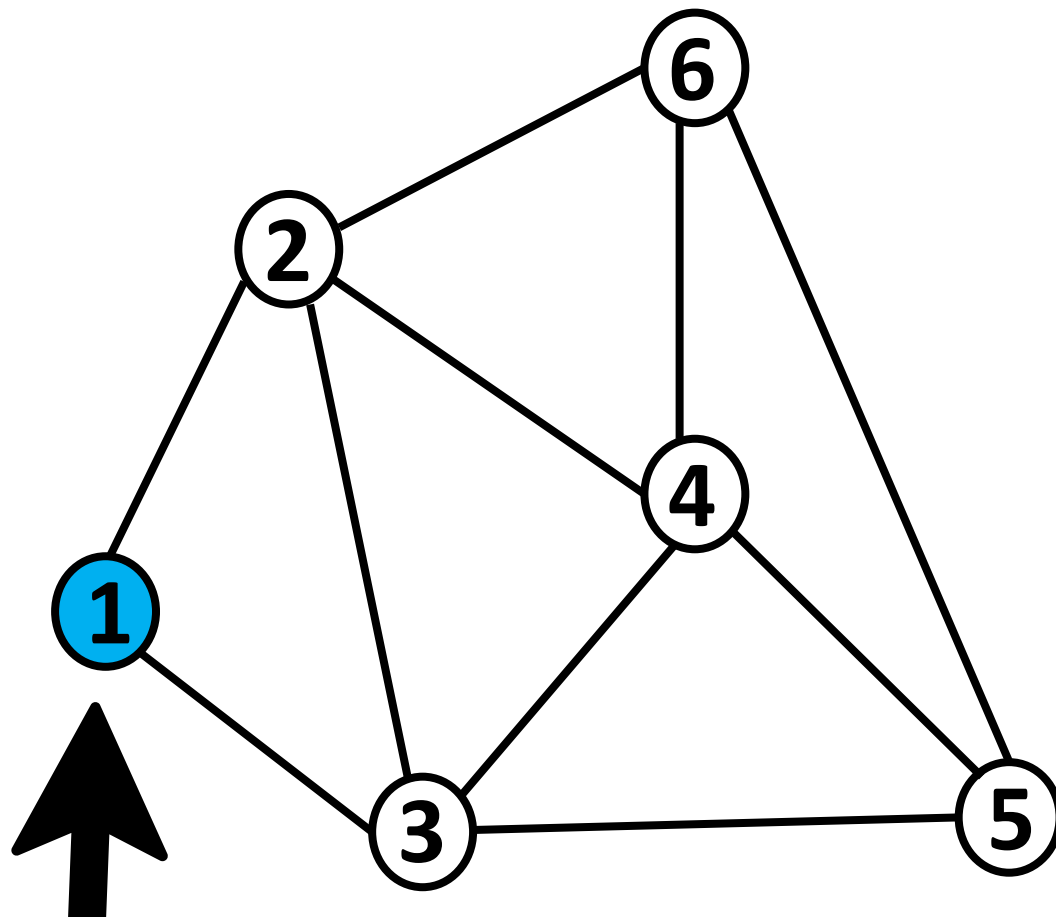


$$x_1 > x_2 > \dots > x_6$$

Graph Colouring

$$x_i \neq x_j$$

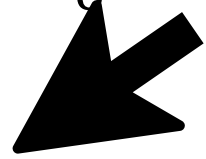
$$\begin{aligned} x_1 &= \{ \text{blue circle} \text{ } \text{orange circle} \text{ } \text{green circle} \} \\ x_2 &= \{ \text{blue circle} \text{ } \text{orange circle} \text{ } \text{green circle} \} \\ x_3 &= \{ \text{blue circle} \text{ } \text{orange circle} \text{ } \text{green circle} \} \\ x_4 &= \{ \text{blue circle} \text{ } \text{orange circle} \text{ } \text{green circle} \} \\ x_5 &= \{ \text{blue circle} \text{ } \text{orange circle} \text{ } \text{green circle} \} \\ x_6 &= \{ \text{blue circle} \text{ } \text{orange circle} \text{ } \text{green circle} \} \end{aligned}$$





$$x_1 > x_2 > \dots > x_6$$

$$x_1 = \text{blue circle} @ 1$$

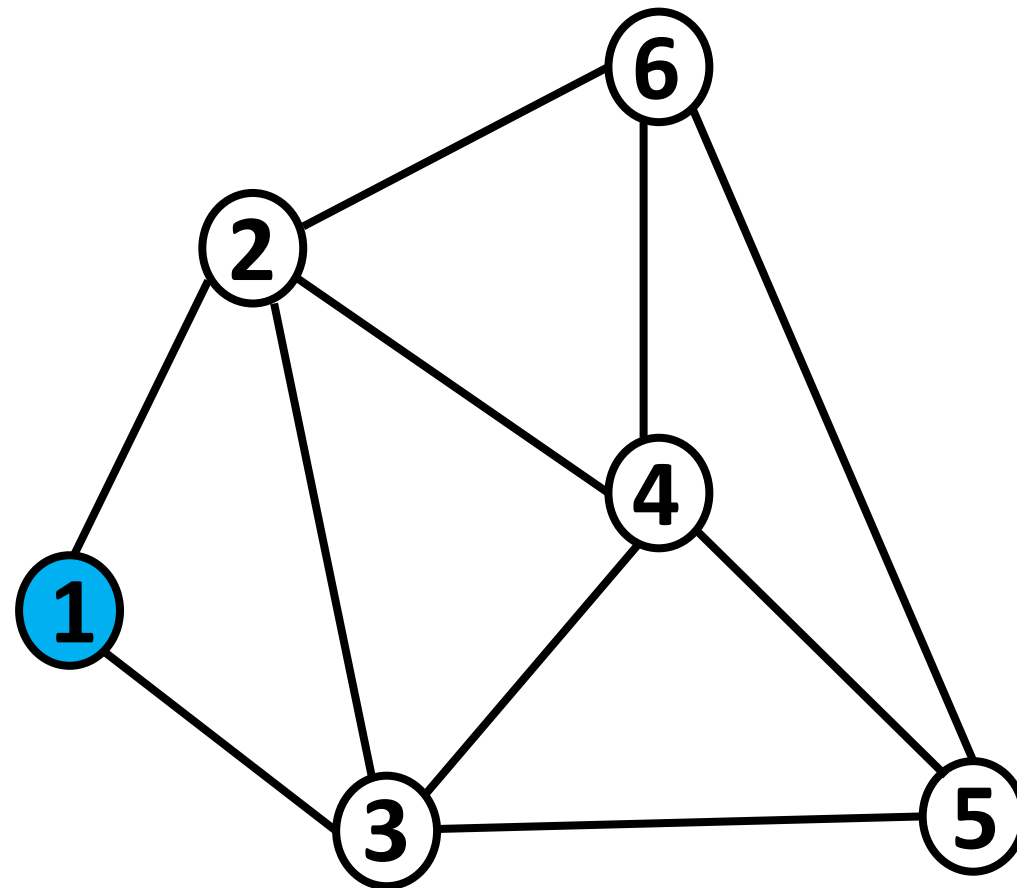


Graph Colouring

$$x_i \neq x_j$$

decision

$x_1 = \{$				$\}$
$x_2 = \{$				$\}$
$x_3 = \{$				$\}$
$x_4 = \{$				$\}$
$x_5 = \{$				$\}$
$x_6 = \{$				$\}$

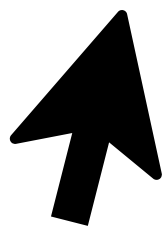




$$x_1 > x_2 > \dots > x_6$$

$$x_1 = \text{blue circle} @ 1$$

$x_2 \neq \text{blue circle} @ 1$
 $x_3 \neq \text{blue circle} @ 1$

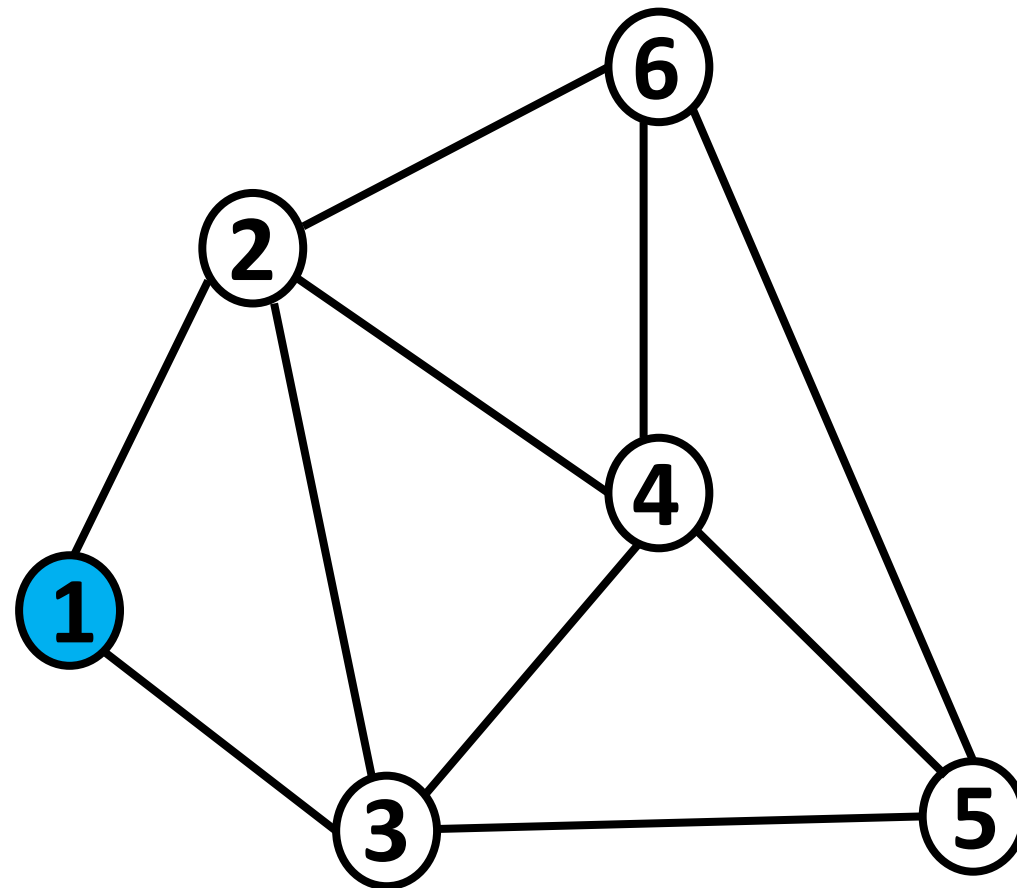


propagations

Graph Colouring

$$x_i \neq x_j$$

$x_1 = \{$				$\}$
$x_2 = \{$				$\}$
$x_3 = \{$				$\}$
$x_4 = \{$				$\}$
$x_5 = \{$				$\}$
$x_6 = \{$				$\}$





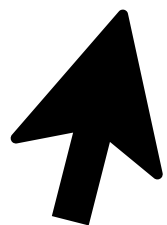
$$x_1 > x_2 > \dots > x_6$$

$$x_1 = \text{blue circle} @ 1$$

$$x_2 \neq \text{blue circle} @ 1$$

$$x_3 \neq \text{blue circle} @ 1$$

$$x_2 = \text{orange circle} @ 2$$

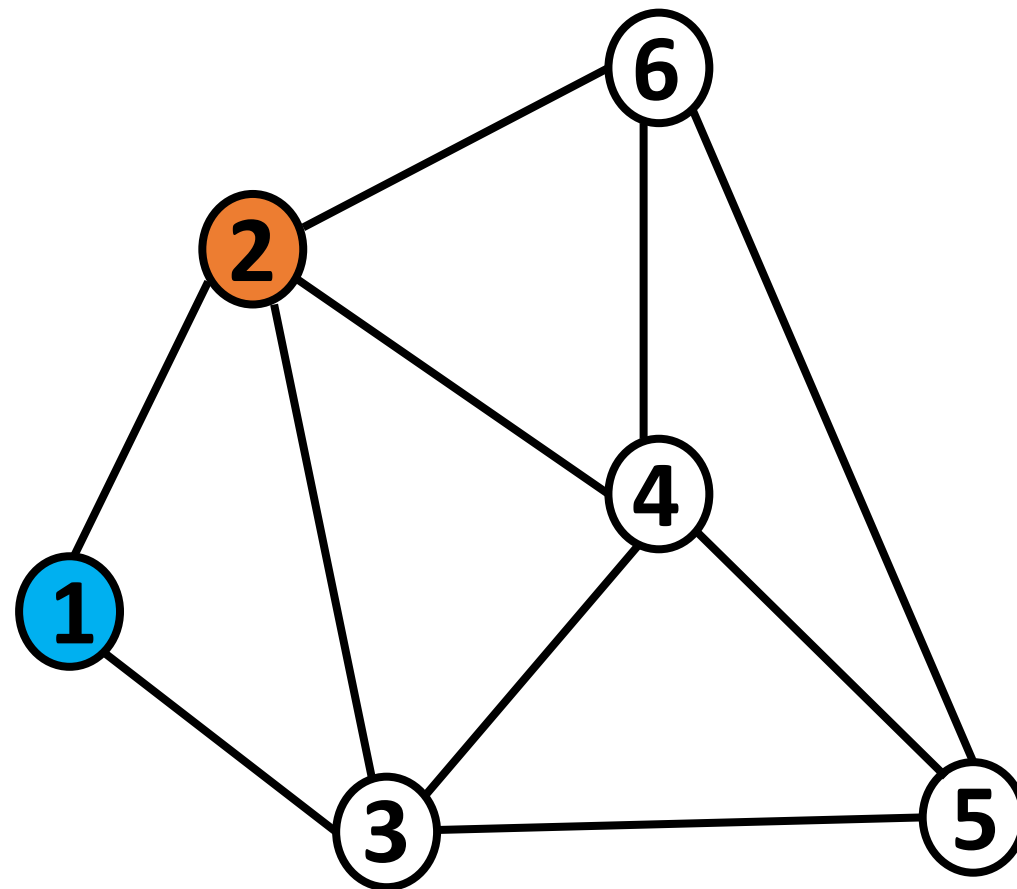


decision

Graph Colouring

$$x_i \neq x_j$$

$$\begin{aligned} x_1 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_2 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_3 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_4 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_5 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \\ x_6 &= \{ \text{blue circle}, \text{orange circle}, \text{green circle} \} \end{aligned}$$





$$x_1 > x_2 > \dots > x_6$$

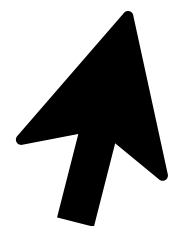
$$x_1 = \text{blue circle} @ 1$$

$$x_2 \neq \text{blue circle} @ 1$$

$$x_3 \neq \text{blue circle} @ 1$$

$$x_2 = \text{orange circle} @ 2$$

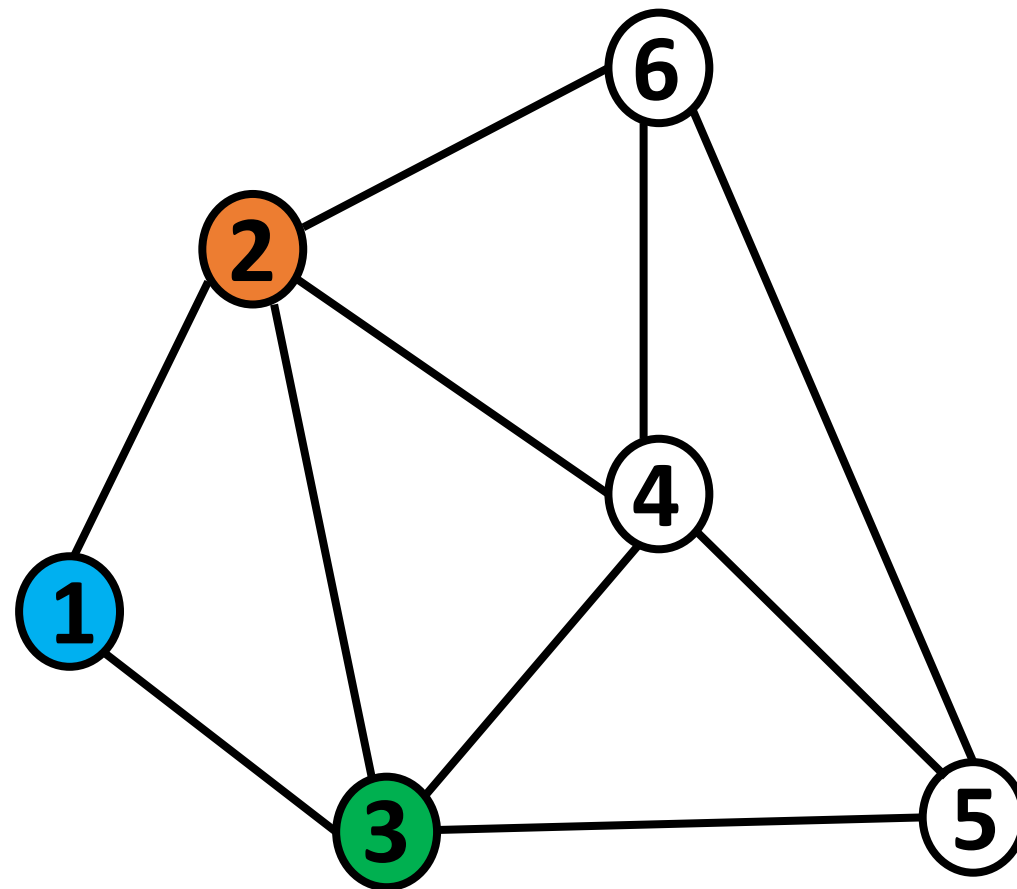
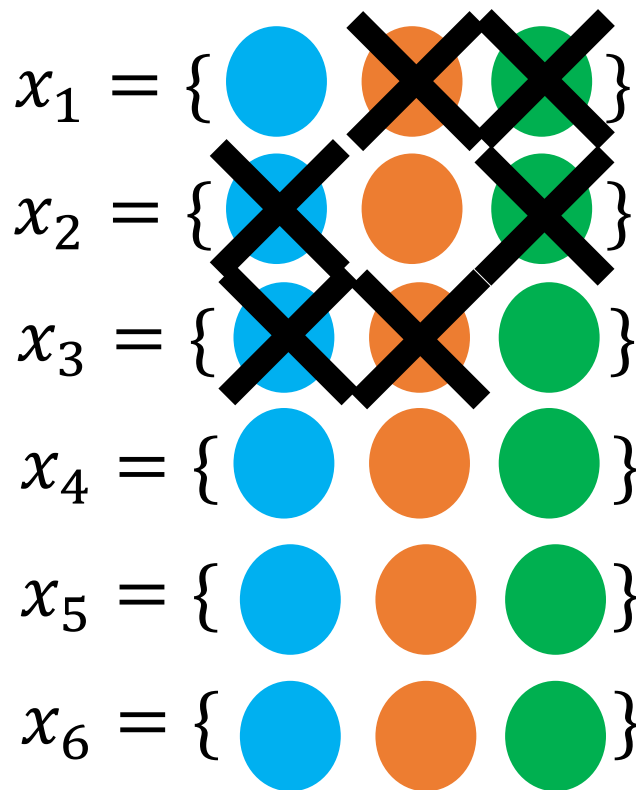
$$x_3 \neq \text{orange circle} @ 2$$



propagation

Graph Colouring

$$x_i \neq x_j$$





$$x_1 > x_2 > \dots > x_6$$

$$x_1 = \text{blue} @ 1$$

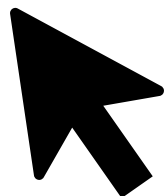
$$x_2 \neq \text{blue} @ 1$$

$$x_3 \neq \text{blue} @ 1$$

$$x_2 = \text{orange} @ 2$$

$$x_3 \neq \text{orange} @ 2$$

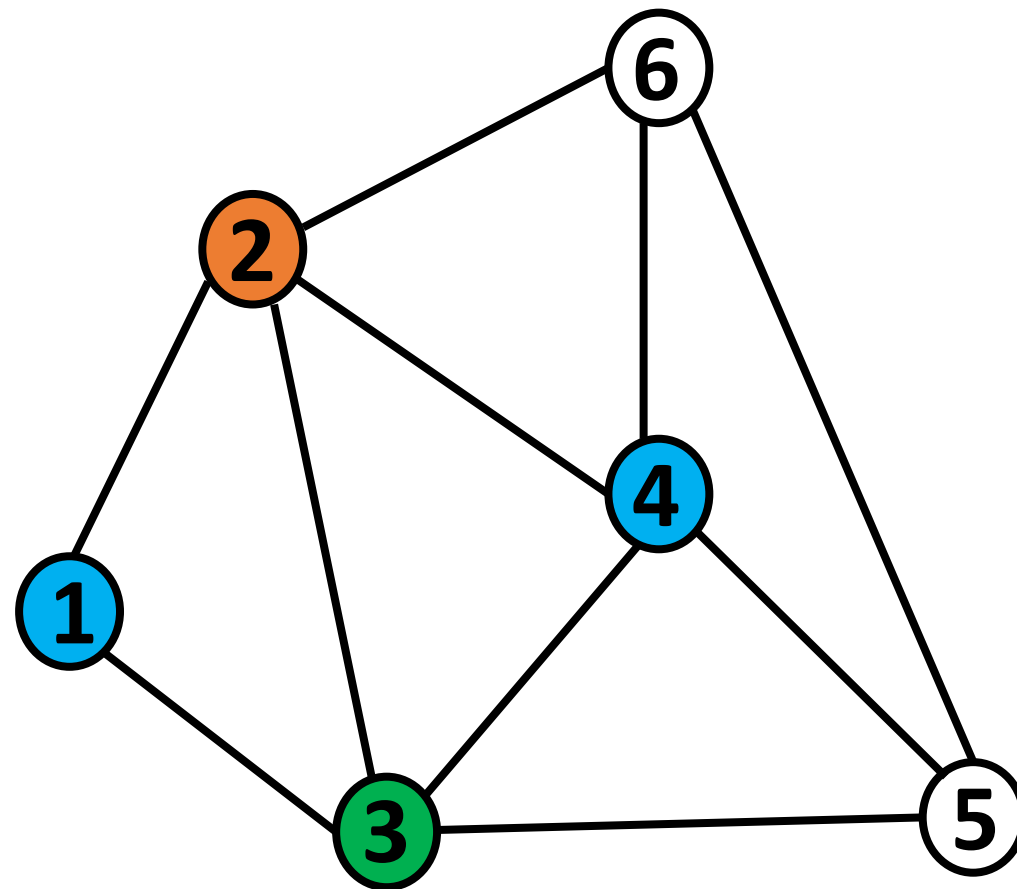
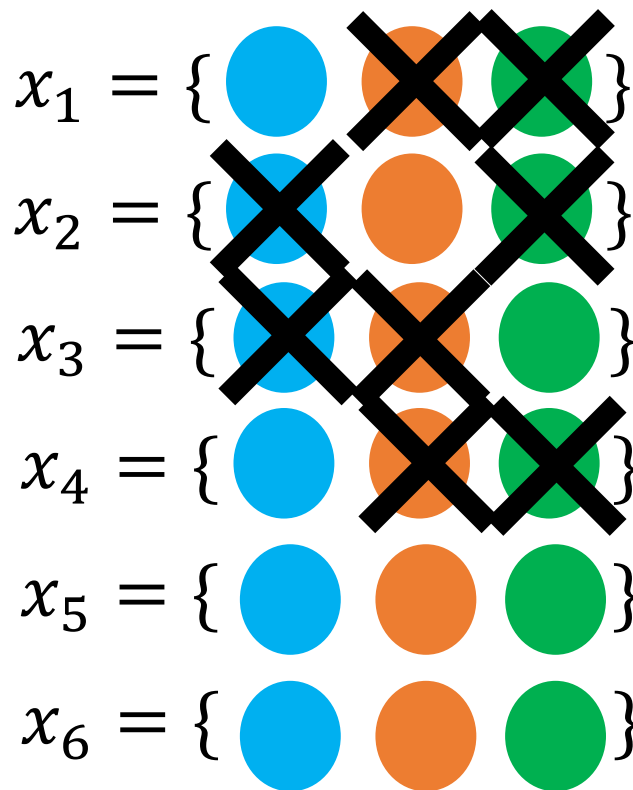
$$x_4 = \text{blue} @ 2$$



propagation

Graph Colouring

$$x_i \neq x_j$$





$$x_1 > x_2 > \dots > x_6$$

$$x_1 = \text{Blue} @ 1$$

$$x_2 \neq \text{Blue} @ 1$$

$$x_3 \neq \text{Blue} @ 1$$

$$x_2 = \text{Orange} @ 2$$

$$x_3 \neq \text{Orange} @ 2$$

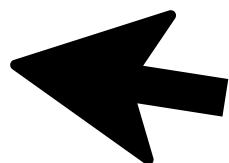
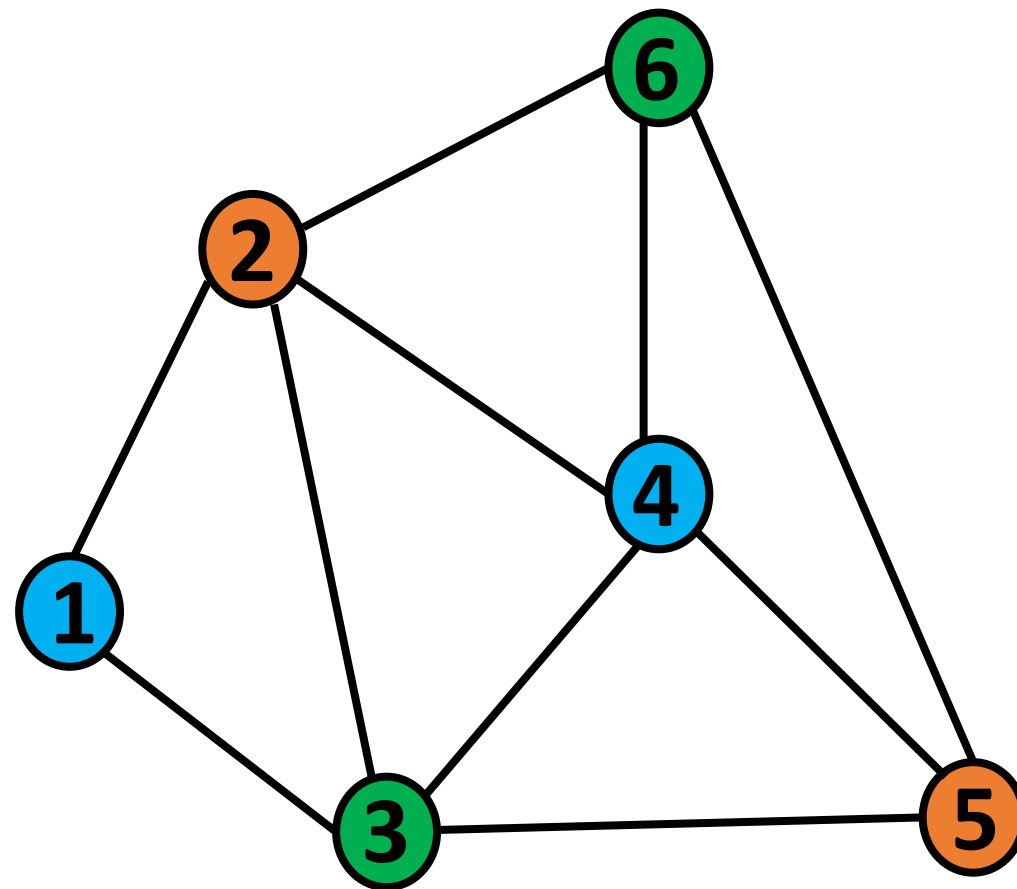
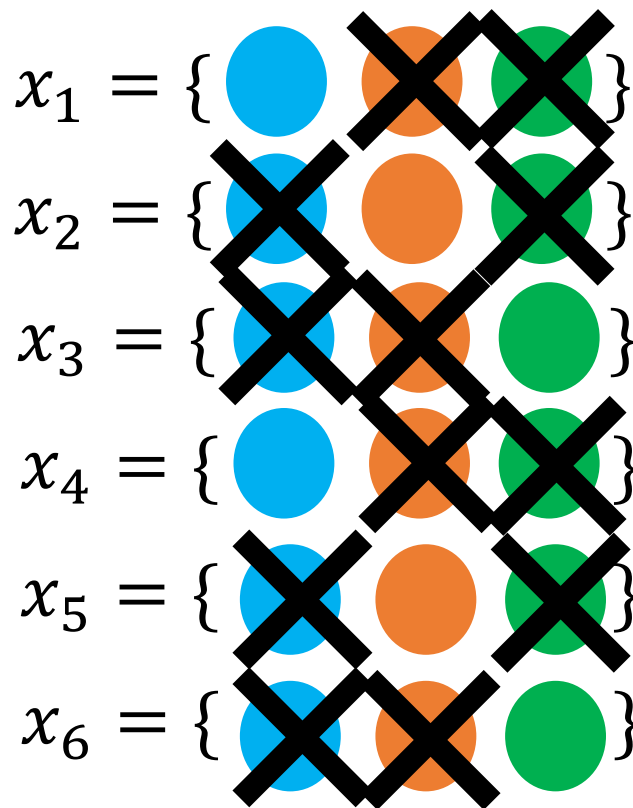
$$x_4 = \text{Blue} @ 2$$

$$x_5 = \text{Orange} @ 2$$

$$x_6 = \text{Green} @ 2$$

Graph Colouring

$$x_i \neq x_j$$



propagation

Problem solved after two decisions!

Search Outline

Make a decision (branch)

Propagate based on constraints

Backtrack if conflict

Constraint Satisfaction Problem

Variables

Integer

$$x_i \in \{0, 1, \dots, 10\}$$

Constraints

Predicates

$$C: X^n \rightarrow \{0, 1\}$$

$$C(x_1, x_2, x_3)$$

Boolean/binary variable

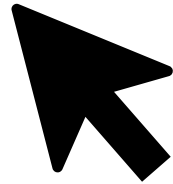
$$x \in \{0, 1\}$$

Integer variable

$$y \in \{0, 1, 2, \dots, n\}$$

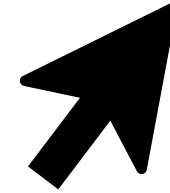
Boolean/binary variable

$$x \in \{0, 1\}$$



Integer variable

$$y \in \{0, 1, 2, \dots, n\}$$



domain

Set of values that the variable
can be potentially assigned

Boolean/binary variable

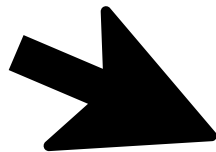
$$x \in \{0, 1\}$$

Integer variable

$$y \in \{0, 1, 2, \dots, n\}$$

domain

Set of values that the variable
can be potentially assigned



We consider finite domains!

Boolean/binary variable

$$x \in \{0, 1\}$$

Integer variable

$$y \in \{0, 1, 2, \dots, n\}$$

Two main representations of the domain

Explicit

$\{0, 1, 5, 6, 7,$
 $10, 11, 12\}$

pros

Allows tracking
“holes” in the domain

cons

Can be
very large

Boolean/binary variable

$$x \in \{0, 1\}$$

Integer variable

$$y \in \{0, 1, 2, \dots, n\}$$

Two main representations of the domain

Explicit

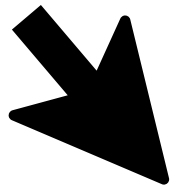
$\{0, 1, 5, 6, 7, 10, 11, 12\}$

pros

Allows tracking
“holes” in the domain

cons

Can be
very large



Interval

$[0, 12]$

Compact regardless
of domain size

“Holes” cannot
be tracked

Boolean/binary variable

$$x \in \{0, 1\}$$

Integer variable

$$y \in \{0, 1, 2, \dots, n\}$$

Two main representations of the domain

Explicit

$\{0, 1, 5, 6, 7, 10, 11, 12\}$

pros

Allows tracking
“holes” in the domain

cons

Can be
very large

Both options used due to unique (dis)advantages

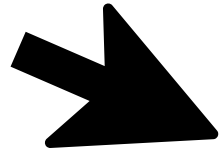
Interval

$[0, 12]$

Compact regardless
of domain size

“Holes” cannot
be tracked

Defines feasibility



Constraint



Propagator



Propagation
Algorithm

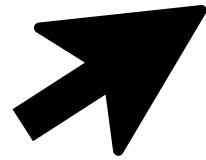
Constraint



Propagator



Propagation
Algorithm



Defines an **inference rule**
to prune (partial) solutions
that cannot be extended to
feasible solutions
based on the constraint

Constraint

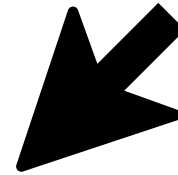


Propagator



Propagation
Algorithm

Defines **how**
the inference is computed



Data structures
extremely
important!

Propagator for the Linear Inequality Constraint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Propagator for the Linear Inequality Constraint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$



Explicit domain representation

$\{0, 2\}$ not possible to express
with interval representation $[0, 2]$

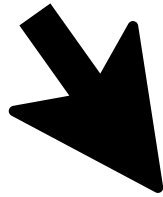
Propagator for the Linear Inequality Constraint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

Propagator



$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\})$$

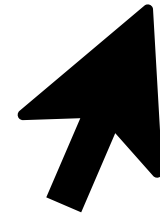
Propagator for the Linear Inequality Constraint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$



No change!

Propagator for the Linear Inequality Constraint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0\}) = (\{4\}, \{0\})$$



Domain changed!

Propagator for the Linear Inequality Constraint

$$y_1 + 2y_2 \geq 4$$

$$y_1 \in \{0, 1, 2, 3, 4\}$$

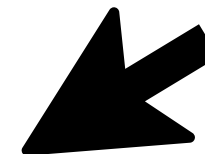
$$y_2 \in \{0, 2\}$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0, 2\}) = (\{0, 1, 2, 3, 4\}, \{0, 2\})$$

$$f(D_1, D_2) = f(\{0, 1, 2, 3, 4\}, \{0\}) = (\{4\}, \{0\})$$

Infeasible, conflict!

$$f(D_1, D_2) = f(\{0, 1, 2, 3, \cancel{4}\}, \{0\}) = (\{\}, \{\})$$



Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Linear Inequality Propagator

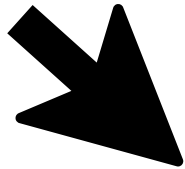
$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

“Upper Bound”



$$y_2 = \textcolor{blue}{UB}(y_2) = 2$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

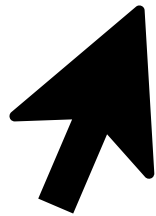
$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = ?$$



Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

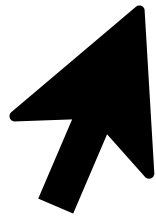
$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$



Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

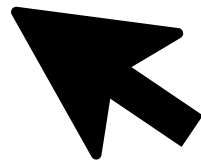
$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$



“Lower Bound”

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

“optimistic assignment”

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = ?$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 16 \geq 17$$


Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = 1$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 16 \geq 17$$


Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{\cancel{0}, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Value '0' removed
by propagation

$$y_2 = UB(y_2) = 2$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_1) = 1$$

$$y_1 + 2 \cdot 2 + 3 \cdot 4 \geq 17$$

$$y_1 + 16 \geq 17$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

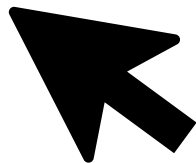
$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$y_1 = UB(y_2) = 4$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_2) = ?$$



Linear Inequality Propagator

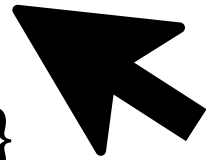
$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{\cancel{0}, 2\}$$

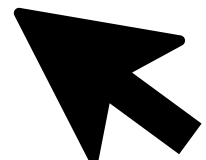
$$y_3 \in \{0, 1, 2, 3, 4\}$$

Value '0' removed
by propagation



$$y_1 = UB(y_2) = 4$$

$$y_3 = UB(y_3) = 4$$

$$LB(y_2) = 2$$


$$4 + 2 \cdot y_2 + 3 \cdot 4 \geq 17$$

$$y_2 + 16 \geq 17$$

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

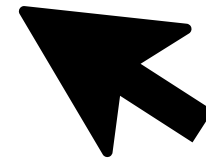
Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{2\}$$

$$y_3 \in \{\cancel{0}, \cancel{1}, \cancel{2}, 3, 4\}$$



Values '0', '1', '2' removed
by propagation

Linear Inequality Propagator

$$y_1 + 2y_2 + 3y_3 \geq 17$$

After
propagation

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{2\}$$

$$y_3 \in \{3, 4\}$$

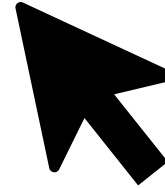
Before
propagation

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

$$f(\{0, 1, 4\}, \{0, 2\}, \{0, 1, 2, 3, 4\}) = (\{1, 4\}, \{2\}, \{3, 4\})$$



After
propagation

$$y_1 \in \{1, 4\}$$

$$y_2 \in \{2\}$$

$$y_3 \in \{3, 4\}$$

Before
propagation

$$y_1 \in \{0, 1, 4\}$$

$$y_2 \in \{0, 2\}$$

$$y_3 \in \{0, 1, 2, 3, 4\}$$

Constraint

$$C: Y^n \rightarrow \{0, 1\}$$

$$C(y_1, y_2, \dots, y_n)$$

$$y_i \in D \subset \mathbb{N}_0$$

Propagator

defines inference for a constraint

$$f: D^n \rightarrow D^n$$

$$f(D_1, D_2, \dots, D_n) = (D'_1, D'_2, \dots, D'_n)$$

$$D_i \subset \mathbb{N}_0, D'_i \subseteq D_i$$

In this mini-lecture...

Model-and-Solve → Constraint Programming

Search

Constraints and Propagators